ASYMPTOTIC SHIFT THRESHOLDS FOR PÓLYA–SCHUR MULTIPLIERS AND A CERTIFIED LOWER BOUND ON THE DE BRUIJN–NEWMAN CONSTANT OF THE DAVENPORT–HEILBRONN FUNCTION

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Abstract. We establish sufficient asymptotic shift conditions for real entire Pólya–Schur multipliers $G(-\partial_z^2)$ acting on the completed Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ and certify explicit unconditional bounds on its de Bruijn–Newman constant Λ_{DH}. Relating $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ via the Stieltjes logarithmic derivative $S_G(u) = \frac{\mathrm{d}}{\mathrm{d}u}\log G(u^2)$ to vertical shifts relative to $\sigma^* := \sup\{\operatorname{Re}\rho : \mathcal{D}_{\mathrm{DH}}(\rho) = 0\}$, we establish sufficient conditions for persistence versus finiteness of non-real zeros, construct tower-lacunary canonical products of every upper order $\rho \in (1/2, 1)$ that preserve infinitely many off-line zeros without root-gap regularity, and prove via a singularity-free interval-arithmetic continuation across $h \in [0, 311/500]$ that Pólya's hyperbolic sinc multiplier strictly separates global real-rootedness from the infinite-height threshold: $\sigma^* - 1/2 < 311/500 < h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \le 31$. Combining complex Interval–Newton off-real zero enclosures under backward heat evolution with an adaptation of the Ki–Kim–Lee upper-bound method yields $1153/5000 < \Lambda_{\mathrm{DH}} < 2(\sigma^* - 1/2)^2$.

1. Introduction

Let χ_5 denote the odd primitive Dirichlet character modulo 5 given by $\chi_5(1) = 1$, $\chi_5(2) = \mathrm{i}$, $\chi_5(3) = -\mathrm{i}$, $\chi_5(4) = -1$, and $\chi_5(5) = 0$, and consider the self-dual Dirichlet combination introduced by Davenport and Heilbronn [7] (see Titchmarsh [26, §10.25]):

$$\kappa := \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}, \qquad \mathcal{D}_{\mathrm{DH}}(s) := \frac{1 - \mathrm{i}\kappa}{2} L(s, \chi_5) + \frac{1 + \mathrm{i}\kappa}{2} L(s, \overline{\chi_5}) = \sum_{n=1}^{\infty} a_{\mathrm{DH}}(n)\, n^{-s}. \tag{1.1}$$

In the canonical normalization of Riemann [22], Pólya [19, 20], de Bruijn [8], and Newman [17], the completed entire function and its theta-kernel cosine representation take the form

$$\begin{aligned} \xi_{\mathrm{DH}}(s) &:= \left(\frac{5}{\pi}\right)^{(s+1)/2} \Gamma\left(\frac{s+1}{2}\right) \mathcal{D}_{\mathrm{DH}}(s), \\ \Xi_{\mathrm{DH}}(z) &:= \xi_{\mathrm{DH}}\left(\tfrac{1}{2} + \tfrac{\mathrm{i}z}{2}\right) = 2\int_0^{\infty} \Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u, \end{aligned} \tag{1.2}$$

where $\Phi_{\mathrm{DH}}(u) := 4\mathrm{e}^{3u}\sum_{n=1}^{\infty} n\, a_{\mathrm{DH}}(n)\, \mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}/5}$ satisfies $\Phi_{\mathrm{DH}}(-u) = \Phi_{\mathrm{DH}}(u)$, encoding $\xi_{\mathrm{DH}}(s) = \xi_{\mathrm{DH}}(1-s)$ and $\Xi_{\mathrm{DH}}(z) = \Xi_{\mathrm{DH}}(-z) = \overline{\Xi_{\mathrm{DH}}(\bar{z})}$. Whereas Riemann's $\Xi(z)$ links critical-line zero geometry to classical criteria of Csordas, Smith, and Varga [6] and satisfies $0 \le \Lambda_\zeta \le 1/5$ [24, 21, 18], $\mathcal{D}_{\mathrm{DH}}(s)$ lacks an Euler product and possesses infinitely many zeros both in $\operatorname{Re}(s) > 1$ and off the critical line inside $0 < \operatorname{Re}(s) < 1$ [7, 25, 1, 4]. Bombieri and Ghosh [4] proved that $\sigma^* := \sup\{\operatorname{Re}\rho : \mathcal{D}_{\mathrm{DH}}(\rho) = 0\}$ is determined via Kronecker's theorem by $\sum_{p\equiv 2,3 \pmod 5} \arctan(p^{-\sigma^*}) = \frac{\pi}{2} - \arctan\kappa$; at the same time, Righetti [23] showed that

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the real parts of zeros of general linear combinations of Euler products in $\mathrm{Re}(s) > 1$ need not be dense in $(1, \sigma^*)$.

Two developments in the de Bruijn–Newman literature directly motivate the analytical framework of the present paper. First, Dobner [9] proved Newman's conjecture for the extended Selberg class by approximating the heat-evolved L-function $\xi_t(s)$ via the modified Dirichlet series $\sum_{n=1}^{\infty} a(n)\, \mathrm{e}^{(t/4)\log^2 n} n^{-s}$; in Section 2 we generalize Dobner's Gaussian multiplier $G_t(w) = \mathrm{e}^{tw}$ to real entire Pólya–Schur multipliers $G(w) = \mathrm{e}^{\lambda w} \prod_{j=1}^{\infty}(1 + w/w_j)$, replacing $\mathrm{e}^{(t/4)\log^2 n}$ by the canonical-product ratio $h_G(\Omega(z) - \log n)/h_G(\Omega(z))$ governed by the Stieltjes logarithmic derivative $S_G(u) = \frac{\mathrm{d}}{\mathrm{d}u} \log G(u^2)$. Second, Ki, Kim, and Lee [13] established the strict upper bound $\Lambda_\zeta < 1/2$ (recently refined for Dirichlet L-functions in [10]) by proving asymptotic simplicity of zeros of H_t at large height and applying de Bruijn's hyperbola inequality on the complementary compact box; in Section 4 we adapt the Ki–Kim–Lee argument to Ξ_{DH} to make the upper bound $\Lambda_{\mathrm{DH}} < 2(\sigma^* - \frac{1}{2})^2$ explicit alongside our certified interval-arithmetic lower bound.

Theorem 1.1. *Let $\mathcal{G}_0$ denote the class of real entire Pólya–Schur multipliers $G(w) = \mathrm{e}^{\lambda w} \prod_{j=1}^{\infty}(1 + w/w_j)$ with $\lambda \geq 0$, $0 < w_1 \leq w_2 \leq \cdots$, and $\sum_{j=1}^{\infty} w_j^{-1} < \infty$, acting via $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = 2\int_0^{\infty} G(u^2)\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u$. Define the Stieltjes logarithmic derivative $S_G(u) := \frac{\mathrm{d}}{\mathrm{d}u}\log G(u^2) = 4\lambda u + \sum_{j=1}^{\infty} \frac{2u}{w_j + u^2}$ $(u > 0)$, and let $N_{\mathrm{off}}(F; T)$ count the non-real zeros of $F(z)$ in $|\mathrm{Re}\, z| \leq 2T$ with multiplicity.*

(i) Sufficient Stieltjes shift conditions and lacunary obstruction.
  (a) *If $\lambda = 0$ and $\sum_{j=1}^{\infty} w_j^{-1/2} < \infty$ (including every polynomial and every multiplier of upper order $\rho(G) < 1/2$), then $\lim_{u\to\infty} S_G(u) = 0$ and $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}}; T) \gg_G T$ inside $|\mathrm{Im}\, z| > 1$. In particular, $(1 - \beta\partial_z^2)\Xi_{\mathrm{DH}}(z)$ has infinitely many non-real zeros for every $\beta > 0$.*
  (b) *If $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = \tau \in (0, \infty)$, then $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}}; T) \gg_G T$ for every $0 < \tau < 2\sigma^* - 1$, whereas $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}}; \infty) < \infty$ for every $\tau > 2\sigma^* - 1$.*
  (c) *Every $G \in \mathcal{G}_0$ with $\lambda > 0$ or lower order $\underline{\rho}(G) > 1/2$ satisfies $\lim_{u\to\infty} S_G(u) = +\infty$. Moreover, $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}}; \infty) < \infty$ whenever $\lambda > 0$, or whenever $\lambda = 0$ and $w_{j+1} - w_j = o(\sqrt{w_j})$ as $j \to \infty$ (in particular for $w_j \sim c\, j^{1/\rho}$ with $\rho \in (1/2, 1)$). Conversely, for every $\rho \in (1/2, 1)$ there exists a lacunary canonical product $G \in \mathcal{G}_0$ of upper order $\rho(G) = \rho$ and lower order $\underline{\rho}(G) = 0$ such that $\liminf_{u\to\infty} S_G(u) = 0$ and $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ has infinitely many non-real zeros in $|\mathrm{Im}\, z| > 1$.*

(ii) Strict separation between asymptotic shift and global real-rootedness. *Comparing $G_{\cosh,h}(w) := \cosh(2h\sqrt{w})$ and Pólya's hyperbolic sinc multiplier $G_{\mathrm{sinc},h}(w) := \frac{\sinh(2h\sqrt{w})}{2h\sqrt{w}}$, both having $\lim_{u\to\infty} S_G(u) = 2h$, the global thresholds $h^*_{\mathrm{global}}(G_\bullet; \Xi_{\mathrm{DH}}) := \inf\{h > 0 : G_{\bullet,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ has only real zeros$\}$ satisfy*

$$h^*_{\mathrm{global}}(G_{\cosh}; \Xi_{\mathrm{DH}}) = \sigma^* - \tfrac{1}{2} < \frac{311}{500} < h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \leq 31. \tag{1.3}$$

(iii) Unconditional bounds on Λ_{DH}. *Under backward heat evolution $H_t^{\mathrm{DH}}(z) := \mathrm{e}^{-t\partial_z^2}\Xi_{\mathrm{DH}}(z)$, the de Bruijn–Newman constant $\Lambda_{\mathrm{DH}} := \inf\{t \in \mathbb{R} : H_t^{\mathrm{DH}}$ has only real zeros$\}$ satisfies*

$$\frac{1153}{5000} < \Lambda_{\mathrm{DH}} < 2\Big(\sigma^* - \tfrac{1}{2}\Big)^2. \tag{1.4}$$

The constants in Theorem 1.1 are certified in interval ball arithmetic (Lemma 3.1) to satisfy

$$\sigma^* = 1.120362498\ldots, \quad h^*_\infty := \sigma^* - \tfrac{1}{2} = 0.620362498\ldots, \quad 2(h^*_\infty)^2 = 0.769699258\ldots \tag{1.5}$$

For comparison, Platt and Trudgian [18] proved unconditionally that $\Lambda_\zeta \leq \frac{1}{5} = 0.2$, so (1.4) implies in particular $\Lambda_{\mathrm{DH}} > \frac{1153}{5000} = 0.2306 > \Lambda_\zeta$.

1.1. **Proof overview.** The proof of Theorem 1.1 rests on three technical components:

(1) *Conjugate-diagram Borel–Laplace asymptotics and single-zero syndetic translation (Section 2):* When $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = \tau_0 \in [0,\infty)$, Pólya's conjugate-diagram representation transfers $G(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ to a compact stadium contour around $[-\frac{\tau_0}{2}, \frac{\tau_0}{2}]$, generalizing Dobner's modified Dirichlet series [9] via $h_G(\Omega(z) - \log n)/h_G(\Omega(z))$. For $0 \le \tau_0 < 2\sigma^* - 1$, translating a single off-line zero in $\mathrm{Re}(s) > 1$ by Dirichlet simultaneous approximation avoids Righetti's non-density phenomenon [23] and yields $\gg_G T$ non-real zeros via Rouché's theorem; for $\tau_0 > 2\sigma^* - 1$, diagonal Bohr compactness on $\prod_p S^1$ and seamless Rouché rectangles along $\mathbb{R}$ exclude non-real zeros above a finite height, while subproduct extraction on disjoint intervals (t_m, t_{m+1}) extends finiteness to $w_{j+1} - w_j = o(\sqrt{w_j})$.

(2) *Singularity-free interval continuation for $G_{\mathrm{sinc},h}$ across $[0, 311/500]$ (Section 3):* Whereas $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ becomes globally real-rooted at $h = \sigma^* - \frac{1}{2}$ by the Hermite–Biehler theorem, rescaling $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ to the even Taylor expansion of $\frac{1}{2}\int_{-1}^{1} \tilde{\xi}_{\mathrm{DH}}(s(z) + hv)\,\mathrm{d}v$ removes the apparent singularity at $h = 0$ and allows a 400-ball Interval–Newton continuation from ρ_{2442} at $h = 0$ all the way to $h = \frac{311}{500}$ independently of Section 2, complemented by an explicit Binet–Stirling proof that $h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \le 31$.

(3) *Ki–Kim–Lee asymptotic simplicity and 448-bit Interval–Newton heat enclosures (Sections 4–5):* Writing $H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) = A_t(\gamma,\delta) + \overline{A_t(\gamma,-\delta)}$ on $[0,\infty)$ and shifting the Gaussian contour by $\mathrm{i}\frac{\pi t}{8}$ yields an explicit height $\gamma_0(t)$ above which all zeros of H_t^{DH} are real and simple, from which the Ki–Kim–Lee hyperbola argument [13, 8] gives Corollary 4.4, while a 448-bit `FLINT/Arb` Interval–Newton contraction with third-order derivative Taylor enclosure near ρ_{240} at $t = \frac{1153}{5000}$ certifies $\Lambda_{\mathrm{DH}} > \frac{1153}{5000}$.

Combining Component (1) with Proposition 2.7 establishes Theorem 1.1(i); Component (2) proves (1.3) independently of Theorem 1.1(i); and Component (3) proves (1.4).

1.2. **Notation and conventions.** Throughout, $s = \sigma + \mathrm{i}\gamma$ ($\sigma, \gamma \in \mathbb{R}$) denotes the Dirichlet variable and $z = x - 2\mathrm{i}\delta = 2\gamma - 2\mathrm{i}\delta$ ($\delta = \sigma - \frac{1}{2}$) denotes the spectral variable in (1.2), mapping $\sigma = \frac{1}{2}$ to $z \in \mathbb{R}$ and $|\sigma - \frac{1}{2}| \le h^*_\infty$ to $|\mathrm{Im}\, z| \le 2h^*_\infty$. Under $\partial_t H_t = -\partial_z^2 H_t$, Λ matches the convention of de Bruijn [8], Newman [17], Rodgers–Tao [24], Polymath [21], and Platt–Trudgian [18]; in the rescaled variable $w = z/2$ with $\widetilde{\Xi}(w) = \xi(\frac{1}{2} + \mathrm{i}w)$, one has $\widetilde{\Lambda} = \Lambda/4$. We write $\Gamma_{\mathrm{DH}}(s) := (5/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})$. Estimates $f \ll_\alpha g$ or $f = \mathcal{O}_\alpha(g)$ denote $|f| \le C(\alpha)|g|$ with $C(\alpha)$ depending only on α. Closed interval balls $[m \pm r]$ denote rigorous enclosures $\{w : |w - m| \le r\}$ computed in `FLINT/Arb` [11].

2. Sufficient Stieltjes Shift Conditions and Root-Gap Obstruction

Let $G(w) = \mathrm{e}^{\lambda w}\prod_{j=1}^{\infty}(1 + w/w_j) \in \mathcal{G}_0$, where $\lambda \ge 0$, $0 < w_1 \le w_2 \le \cdots$, and $\sum_{j=1}^{\infty} w_j^{-1} < \infty$. On every horizontal strip $|\mathrm{Im}\, u| \le \frac{\pi}{8} - \delta_0$ ($\delta_0 \in (0, \frac{\pi}{8})$), the elementary bound $1 + r \le \mathrm{e}^r$ gives $|G(u^2)| \le \mathrm{e}^{(\lambda + \sum_{j=1}^{\infty} w_j^{-1})|u|^2}$, whereas $\mathrm{Re}(\mathrm{e}^{4u}) = \mathrm{e}^{4\,\mathrm{Re}\,u}\cos(4\,\mathrm{Im}\,u) \ge \sin(4\delta_0)\,\mathrm{e}^{4\,\mathrm{Re}\,u}$ implies $|\Phi_{\mathrm{DH}}(u)| \ll_{\delta_0} \mathrm{e}^{3|\mathrm{Re}\,u| - \frac{\pi}{5}\sin(4\delta_0)\mathrm{e}^{4|\mathrm{Re}\,u|}}$. Consequently, the integral

$$G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = 2\int_0^{\infty} G(u^2)\,\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u = \int_{-\infty}^{\infty} G(u^2)\,\Phi_{\mathrm{DH}}(u)\,\mathrm{e}^{\mathrm{i}zu}\,\mathrm{d}u \tag{2.1}$$

converges absolutely and locally uniformly on $\mathbb{C}$ to a real-even entire function of order 1. Because $G(u^2) > 0$ for all $u \in \mathbb{R}$ and Φ_{DH} is continuous and not identically zero, Fourier uniqueness ensures $G(-\partial_z^2)\Xi_{\mathrm{DH}} \not\equiv 0$. Furthermore, for any factorization $G = G^{(1)}G^{(2)}$ in $\mathcal{G}_0$, the finite truncations $G_N^{(2)}(w) := \mathrm{e}^{\lambda_2 w}\prod_{j=1}^{N}(1 + w/w_{2,j})$ satisfy $0 < G_N^{(2)}(u^2) \le G^{(2)}(u^2)$ on $\mathbb{R}$, so dominated convergence in (2.1) yields $G_N^{(2)}(-\partial_z^2)[G^{(1)}(-\partial_z^2)\Xi_{\mathrm{DH}}](z) \to G(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ locally uniformly on $\mathbb{C}$.

We begin by recording two classical zero-preservation lemmas for Laguerre–Pólya differential operators.

Lemma 2.1 (Pólya [19, 20], Benz [2]; cf. Lagarias–Suzuki [14])**.** *Let $F(z) \not\equiv 0$ be a real-even entire function of genus 0 or 1 whose zeros all satisfy $|\operatorname{Im} z| \le 2h_0$ ($h_0 > 0$).*

(i) *For every $G \in \mathcal{G}_0$ such that $G_N(-\partial_z^2)F \to G(-\partial_z^2)F \not\equiv 0$ locally uniformly on $\mathbb{C}$, all zeros of $G(-\partial_z^2)F(z)$ lie in the strip $|\operatorname{Im} z| \le 2h_0$.*

(ii) *For $G_{\cosh,h}(w) = \cosh(2h\sqrt{w})$, all zeros of $G_{\cosh,h}(-\partial_z^2)F(z) = \frac{1}{2}[F(z-2\mathrm{i}h)+F(z+2\mathrm{i}h)]$ are real whenever $h \ge h_0$, and real and simple whenever $h > h_0$.*

Proof. For (i), let $H(z) = c\,z^m \mathrm{e}^{az}\prod_n (1 - z/\zeta_n)\mathrm{e}^{\mu_n z/\zeta_n}$ ($c \in \mathbb{R}\setminus\{0\}$, $a \in \mathbb{R}$, $\mu_n \in \{0,1\}$, $\sum_{\zeta_n\neq 0}|\zeta_n|^{-2} < \infty$) be any real entire function of genus 0 or 1 whose zeros satisfy $|\operatorname{Im}\zeta_n| \le 2h_0$. Grouping non-real zeros into conjugate pairs $\{\zeta_n, \overline{\zeta_n}\}$ cancels $\operatorname{Im}(\mu_n/\zeta_n + \mu_n/\overline{\zeta_n}) = 0$, so taking imaginary parts of $H'(z_0)/H(z_0)$ gives the absolutely convergent sum $\operatorname{Im}(H'(z_0)/H(z_0)) = -\frac{m\operatorname{Im} z_0}{|z_0|^2} - \sum_n \frac{\operatorname{Im} z_0 - \operatorname{Im}\zeta_n}{|z_0-\zeta_n|^2} < 0$ for every $\operatorname{Im} z_0 > 2h_0$, whence $(1 \pm \alpha\partial_z)H(z_0) \neq 0$ for all $\alpha \in \mathbb{R}$. Iterating across the linear factors of $G_N(-\partial_z^2)F$ (with de Bruijn [8, Thm. 13] for $\mathrm{e}^{-\lambda\partial_z^2}$) and passing to the limit via Hurwitz's theorem gives (i). For (ii), all zeros of $\omega(z) := F(z + 2\mathrm{i}h)$ lie in $\operatorname{Im} z \le 2(h_0 - h) \le 0$, so factorwise comparison gives $|\omega(z)| > |\overline{\omega(\bar z)}| = |F(z - 2\mathrm{i}h)|$ on $\operatorname{Im} z > 0$, and the Hermite–Biehler theorem yields simplicity on $\mathbb{R}$ when $h > h_0$. □

Lemma 2.2 (Non-real zero monotonicity; cf. Pólya [20], Craven–Csordas–Smith [5])**.** *Let $F(z)$ be a real entire function of order at most 1 with $N_{\mathrm{off}}(F;\infty) = 2M < \infty$. Then for every $G \in \mathcal{G}_0$ such that $G_N(-\partial_z^2)F \to G(-\partial_z^2)F \not\equiv 0$ locally uniformly on $\mathbb{C}$, one has $N_{\mathrm{off}}(G(-\partial_z^2)F;\infty) \le 2M$.*

Proof. Factor $F = P_{2M}E$, where $\deg P_{2M} = 2M$ and E lies in the Laguerre–Pólya class of order at most 1. Approximating E locally uniformly by real-rooted polynomials Q_K of degree K, the polynomial $R_K := P_{2M}Q_K$ of degree $2M + K$ has at least K real roots counted with multiplicity. For any $\alpha \in \mathbb{R}\setminus\{0\}$, Rolle's theorem applied to $\mathrm{e}^{x/\alpha}R_K(x)$ between consecutive real roots and along the half-line where $\mathrm{e}^{x/\alpha}R_K(x) \to 0$ produces at least K real roots of $(1 + \alpha\partial_z)R_K$, leaving at most $2M$ non-real roots. Letting $K \to \infty$ and $G_N(-\partial_z^2)F \to G(-\partial_z^2)F$ via Hurwitz's theorem preserves $N_{\mathrm{off}} \le 2M$. □

Remark 2.3. For the single derivative $F \mapsto F'$, the refined asymptotic $N_{\mathrm{off}}(F;T) - N_{\mathrm{off}}(F';T) = 2D(F;T) + \mathcal{O}(\log T)$ counting real critical points of F/F' was proved for Riemann's Ξ by Levinson and Montgomery [16] and for real entire functions of genus 0 or 1 by Ki and Kim [12]; here only the global finiteness bound of Lemma 2.2 is required.

We next isolate the vertical-strip growth, single-zero Diophantine translation, and supercritical lower-bound properties of $\mathcal{D}_{\mathrm{DH}}(s)$ and $\xi_{\mathrm{DH}}(s)$ needed in the contour analysis.

Lemma 2.4 (Vertical-strip convexity, syndetic zero translation, and supercritical lower bound)**.**

(i) *For every $\eta \in (0, \frac{1}{2})$ and $A \ge 2$, there exists a constant $C_0(\eta, A) > 0$ such that for all $\sigma \in [-A, A]$ and all $\tau \ge 2$,*

$$|\xi_{\mathrm{DH}}(\sigma + \mathrm{i}\tau)| \le C_0(\eta, A)\,\tau^{\frac{1}{2}\max(\sigma,\,1-\sigma,\,1+\eta)}\,\mathrm{e}^{-\frac{\pi}{4}\tau}. \tag{2.2}$$

(ii) *For every $\sigma_{\mathrm{low}} \in [1, \sigma^*)$, there exist a zero $\rho_* = \beta_0 + \mathrm{i}\gamma_*$ of $\mathcal{D}_{\mathrm{DH}}(s)$ with $\beta_0 \in (\sigma_{\mathrm{low}}, \sigma^*]$, radii $\eta_0 := \frac{1}{4}(\beta_0 - \sigma_{\mathrm{low}}) > 0$ and $r_0 \in (0, \min(\eta_0/2, 1/4))$, a constant $\mu_0 > 0$, and a threshold $T_0 \ge 4$ such that for every $T \ge T_0$ there exist $M(T) \gg_{\sigma_{\mathrm{low}}} T$ real shifts $\tau_1 < \tau_2 < \cdots < \tau_{M(T)}$ in $[T, 2T]$ satisfying $\tau_{m+1} - \tau_m \ge 1$ such that on every translated circle $|w - (\rho_* + \mathrm{i}\tau_m)| = r_0$,*

$$|\mathcal{D}_{\mathrm{DH}}(w) - \mathcal{D}_{\mathrm{DH}}(w - \mathrm{i}\tau_m)| < \frac{\mu_0}{2}, \qquad |\mathcal{D}_{\mathrm{DH}}(w)| > \frac{\mu_0}{2}, \tag{2.3}$$

and $\mathcal{D}_{\mathrm{DH}}(w)$ has at least one zero inside each disjoint disk $|w - (\rho_* + \mathrm{i}\tau_m)| < r_0$.

(iii) *For every $\varepsilon > 0$, there exists a constant $c(\varepsilon) > 0$ such that $|\mathcal{D}_{\mathrm{DH}}(w)| \geq c(\varepsilon)$ for all $w \in \mathbb{C}$ with $\operatorname{Re} w \geq \sigma^* + \varepsilon$.*

Proof. (i) For $\sigma \in [1+\eta, A]$ and $\tau \geq 2$, $|\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)| \leq \zeta(1+\eta)$, while Stirling's formula gives $|\Gamma_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)| \ll_A \tau^{\sigma/2}\mathrm{e}^{-\frac{\pi}{4}\tau}$, establishing (2.2) on $[1+\eta, A]$ since $\max(\sigma, 1-\sigma, 1+\eta) = \sigma$. By the functional equation $|\xi_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)| = |\xi_{\mathrm{DH}}(1-\sigma-\mathrm{i}\tau)| = |\xi_{\mathrm{DH}}(1-\sigma+\mathrm{i}\tau)|$, replacing A by $A+1$ yields $|\xi_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)| \ll_{\eta,A} \tau^{(1-\sigma)/2}\mathrm{e}^{-\frac{\pi}{4}\tau}$ for $\sigma \in [-A, -\eta]$, where $\max(\sigma, 1-\sigma, 1+\eta) = 1-\sigma$. Finally, on the strip $\sigma \in [-\eta, 1+\eta]$, $\mathcal{D}_{\mathrm{DH}}(w)$ has finite order, satisfies $|\mathcal{D}_{\mathrm{DH}}(1+\eta+\mathrm{i}\tau)| \leq \zeta(1+\eta)$ and $|\mathcal{D}_{\mathrm{DH}}(-\eta+\mathrm{i}\tau)| = |\Gamma_{\mathrm{DH}}(1+\eta-\mathrm{i}\tau)/\Gamma_{\mathrm{DH}}(-\eta+\mathrm{i}\tau)|\,|\mathcal{D}_{\mathrm{DH}}(1+\eta-\mathrm{i}\tau)| \ll_\eta \tau^{\frac{1}{2}+\eta}$ for $\tau \geq 2$, so the Phragmén–Lindelöf principle gives $|\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)| \ll_\eta \tau^{\frac{1}{2}(1+\eta-\sigma)}$; multiplying by $|\Gamma_{\mathrm{DH}}(\sigma+\mathrm{i}\tau)| \ll_\eta \tau^{\sigma/2}\mathrm{e}^{-\frac{\pi}{4}\tau}$ yields exponent $\frac{1}{2}(1+\eta) = \frac{1}{2}\max(\sigma, 1-\sigma, 1+\eta)$.

(ii) Because Righetti [23, Thm. 1.2] showed that the real parts of zeros of general linear combinations of Euler products in $\operatorname{Re}(s) > 1$ need not be dense in $(1, \sigma^*)$, one cannot prescribe an arbitrary abscissa in $(\sigma_{\mathrm{low}}, \sigma^*)$. Instead, by the supremum definition $\sigma^* = \sup\{\operatorname{Re}\rho : \mathcal{D}_{\mathrm{DH}}(\rho) = 0\} > \sigma_{\mathrm{low}}$, we fix a single existing zero $\rho_* = \beta_0 + \mathrm{i}\gamma_*$ of $\mathcal{D}_{\mathrm{DH}}(s)$ with $\beta_0 \in (\sigma_{\mathrm{low}}, \sigma^*]$ (and multiplicity $m_* \geq 1$). Set $\eta_0 := \frac{1}{4}(\beta_0 - \sigma_{\mathrm{low}}) > 0$, so $\beta_0 \geq 1 + 4\eta_0$. Since the zeros of $\mathcal{D}_{\mathrm{DH}} \not\equiv 0$ are isolated, there exists $r_0 \in (0, \min(\eta_0/2, 1/4))$ such that $\mathcal{D}_{\mathrm{DH}}(w) \neq 0$ for $0 < |w-\rho_*| \leq r_0$, and compactness gives $\mu_0 := \min_{|w-\rho_*|=r_0} |\mathcal{D}_{\mathrm{DH}}(w)| > 0$. Every w_0 with $|w_0-\rho_*| = r_0$ satisfies $\operatorname{Re} w_0 \geq \beta_0 - r_0 > 1+3\eta_0$. Choose $N_0 \geq 2$ such that $\sum_{n>N_0} n^{-(1+3\eta_0)} < \mu_0/6$, and then choose $\delta_{\mathrm{D}} \in (0, \frac{1}{4})$ such that $2\pi\delta_{\mathrm{D}} \sum_{n=2}^{N_0} n^{-(1+3\eta_0)} < \mu_0/6$. Let $p_1 < \cdots < p_r$ be the primes $\leq N_0$, set $L_0 := \log_2 N_0$ and $Q := \lceil L_0/\delta_{\mathrm{D}} \rceil \geq 2$, and partition the r-torus $(\mathbb{R}/\mathbb{Z})^r$ into $K := (2Q)^r$ half-open subcubes $C_1, \ldots, C_K$ of ℓ^∞-diameter $\frac{1}{2Q}$. For each index ν such that $\{u \geq 0 : u\alpha \bmod 1 \in C_\nu\} \neq \emptyset$, where $\alpha := (\frac{\log p_1}{2\pi}, \ldots, \frac{\log p_r}{2\pi})$, pick one witness $u_\nu \geq 0$ and set $U_0 := 1 + \max_\nu u_\nu < \infty$. For any $X \geq U_0$, the point $X\alpha \bmod 1$ lies in some non-empty C_ν, so $\tau(X) := X - u_\nu \in [X - U_0, X]$ satisfies $\max_{1\leq\ell\leq r} \operatorname{dist}(\tau(X)\frac{\log p_\ell}{2\pi}, \mathbb{Z}) \leq \frac{1}{2Q}$. Setting $X_m := T + m(U_0+1)$ for $1 \leq m \leq \lfloor \frac{T}{U_0+1} \rfloor$ ($T \geq T_0 := 2U_0+4$) produces $M(T) \gg_{\sigma_{\mathrm{low}}} T$ shifts $\tau_m := \tau(X_m) \in [T, 2T]$ with $\tau_{m+1} - \tau_m \geq 1$. For every $2 \leq n = \prod_{\ell=1}^r p_\ell^{\nu_\ell(n)} \leq N_0$, $\sum_{\ell=1}^r \nu_\ell(n) \leq L_0$ implies $\operatorname{dist}(\tau_m \frac{\log n}{2\pi}, \mathbb{Z}) \leq \frac{L_0}{2Q} < \delta_{\mathrm{D}}$ and $|n^{-\mathrm{i}\tau_m} - 1| \leq 2\pi\delta_{\mathrm{D}}$. Hence for $|w_0 - \rho_*| = r_0$,

$$|\mathcal{D}_{\mathrm{DH}}(w_0 + \mathrm{i}\tau_m) - \mathcal{D}_{\mathrm{DH}}(w_0)| \leq \sum_{n=2}^{N_0} n^{-(1+3\eta_0)}|n^{-\mathrm{i}\tau_m} - 1| + 2\sum_{n>N_0} n^{-(1+3\eta_0)} < \frac{\mu_0}{6} + \frac{\mu_0}{3} = \frac{\mu_0}{2},$$

giving (2.3), and Rouché's theorem produces $m_* \geq 1$ zeros inside $|w - (\rho_* + \mathrm{i}\tau_m)| < r_0$.

(iii) For $\operatorname{Re} w \geq 3$, $|\mathcal{D}_{\mathrm{DH}}(w)| \geq 1 - \sum_{n=2}^\infty n^{-3} = 2 - \zeta(3) > \frac{3}{4}$. Suppose toward a contradiction that $\mathcal{D}_{\mathrm{DH}}(\sigma_k + \mathrm{i}\tau_k) \to 0$ for a sequence with $\sigma_k \in [\sigma^* + \varepsilon, 3]$ and $\tau_k \in \mathbb{R}$. Passing to a diagonal subsequence, we have $\sigma_k \to \sigma_\infty \in [\sigma^* + \varepsilon, 3]$ and $p^{-\mathrm{i}\tau_k} \to \omega(p) \in S^1$ for every prime p; extending ω multiplicatively to $\mathbb{N}$, dominated convergence on $\sum_{n=1}^\infty n^{-(\sigma^*+\varepsilon)} < \infty$ yields $\frac{1-\mathrm{i}\kappa}{2} L_\omega(\sigma_\infty, \chi_5) + \frac{1+\mathrm{i}\kappa}{2} L_\omega(\sigma_\infty, \overline{\chi_5}) = \lim_{k\to\infty} \mathcal{D}_{\mathrm{DH}}(\sigma_k + \mathrm{i}\tau_k) = 0$, where $L_\omega(s, \chi) := \prod_p (1 - \chi(p)\omega(p)p^{-s})^{-1} \neq 0$. Thus $L_\omega(\sigma_\infty, \overline{\chi_5})/L_\omega(\sigma_\infty, \chi_5) = -\frac{1-\mathrm{i}\kappa}{1+\mathrm{i}\kappa} = \mathrm{e}^{+2\mathrm{i}(\frac{\pi}{2} - \arctan\kappa)}$. In this Euler product ratio, primes $p \equiv 1, 4 \pmod 5$ have real $\chi_5(p) \in \{\pm 1\}$ and cancel identically for every ω, while each $p \equiv 2, 3 \pmod 5$ contributes $\frac{1-u_p}{1+u_p}$ with $|u_p| = p^{-\sigma_\infty}$, whose principal argument is bounded in modulus by $2\arctan(p^{-\sigma_\infty})$. Since $\sigma_\infty \geq \sigma^* + \varepsilon > \sigma^*$,

$$\begin{aligned}\left|\sum_{p\equiv 2,3 \pmod 5} \arg\frac{1-u_p}{1+u_p}\right| &\leq \sum_{p\equiv 2,3 \pmod 5} 2\arctan(p^{-\sigma_\infty}) \\ &< \sum_{p\equiv 2,3 \pmod 5} 2\arctan(p^{-\sigma^*}) = 2\Big(\frac{\pi}{2} - \arctan\kappa\Big) < \pi,\end{aligned}$$

which prevents equality modulo 2π, a contradiction. □

We now establish the quantitative Borel–Laplace cap formula generalizing Dobner's modified Dirichlet-series approximation [9, Lem. 2.3] from the Gaussian multiplier $G_t(w) = \mathrm{e}^{tw}$ to any $G \in \mathcal{G}_0$ whose Stieltjes derivative converges to a finite limit.

Lemma 2.5 (Quantitative Borel–Laplace cap approximation; cf. Dobner [9])**.** *Let $G \in \mathcal{G}_0$ satisfy $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = \tau_0 \in [0,\infty)$, and define the even entire function $h_G(w) := G(\frac{1}{4}w^2) = \prod_{j=1}^{\infty}(1 + \frac{w^2}{4w_j})$.*

(i) *As $|w| \to \infty$ within the sector $|\arg w| \le \frac{\pi}{3}$, one has $\frac{h_G'(w)}{h_G(w)} \to \frac{\tau_0}{2}$ uniformly, and the Borel transform $\mathcal{B}_G(\zeta)$ of h_G is holomorphic on $\mathbb{C} \setminus [-\frac{\tau_0}{2}, \frac{\tau_0}{2}]$ with $\mathcal{B}_G(-\zeta) = -\mathcal{B}_G(\zeta)$. Moreover, for every $\varepsilon > 0$ and counterclockwise stadium contour $\Gamma_{\tau_0,\varepsilon}$ at distance ε around $[-\frac{\tau_0}{2}, \frac{\tau_0}{2}]$,*

$$F_G(z) := G(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \frac{1}{2\pi\mathrm{i}} \oint_{\Gamma_{\tau_0,\varepsilon}} \mathcal{B}_G(\zeta)\, \xi_{\mathrm{DH}}(s(z)+\zeta)\, \mathrm{d}\zeta. \tag{2.4}$$

(ii) *Fix $\eta \in (0, \frac{1}{16})$, set $\varepsilon := \frac{\eta}{16}$, and let $\Gamma_+ := \{\zeta \in \Gamma_{\tau_0,\varepsilon} : \mathrm{Re}\,\zeta \ge \frac{\tau_0}{2} - \eta\}$. Define $\Omega(z) := \frac{1}{2}\log\frac{5(s(z)+1)}{2\pi}$ (principal branch on $\mathrm{Im}\, s(z) > 0$). Then uniformly for all $z = 2\gamma - 2\mathrm{i}\delta$ with $|\delta| \le 2$ and $\mathrm{Re}\, s(z) + \frac{\tau_0}{2} \ge 1 + 3\eta$, as $\gamma \to +\infty$,*

$$I_+(z) := \frac{1}{2\pi\mathrm{i}} \int_{\Gamma_+} \mathcal{B}_G(\zeta)\, \xi_{\mathrm{DH}}(s(z)+\zeta)\, \mathrm{d}\zeta = \Gamma_{\mathrm{DH}}(s(z))\, h_G(\Omega(z)) \left[\mathcal{D}_{\mathrm{DH}}\big(s(z) + \tfrac{\tau_0}{2}\big) + r_+(z)\right], \tag{2.5}$$

where $\sup\{|r_+(z)| : |\delta| \le 2,\ \mathrm{Re}\, s(z) + \frac{\tau_0}{2} \ge 1 + 3\eta,\ \gamma \ge \Gamma\} \to 0$ as $\Gamma \to \infty$.

Proof. (i) For $\mathrm{Re}\, w > 0$, $h_G(w) \neq 0$ and $\frac{h_G'(w)}{h_G(w)} = \sum_{j=1}^{\infty} \frac{2w}{4w_j + w^2} = \frac{1}{2} S_G(\frac{w}{2})$. Because S_G is continuous on $[0,\infty)$ with $S_G(0) = 0$ and $\lim_{u\to\infty} S_G(u) = \tau_0$, we have $M_G := \sup_{u\ge 0} S_G(u) < \infty$. On the sector $|\arg\zeta| \le \frac{3\pi}{8}$, $|4w_j + R^2\zeta^2| \ge \sin(\frac{\pi}{8})(4w_j + R^2|\zeta|^2)$ for every $R > 0$, so the holomorphic family $q_R(\zeta) := \frac{h_G'(R\zeta)}{h_G(R\zeta)}$ satisfies $|q_R(\zeta)| \le \frac{1}{2\sin(\pi/8)} S_G(\frac{R|\zeta|}{2}) \le \frac{M_G}{2\sin(\pi/8)}$ on $\{\frac{1}{2} \le |\zeta| \le 2,\ |\arg\zeta| \le \frac{3\pi}{8}\}$. Since $\lim_{R\to\infty} q_R(x) = \frac{\tau_0}{2}$ for every real $x \in [\frac{1}{2}, 2]$, Vitali's normal-family theorem implies $q_R(\zeta) \to \frac{\tau_0}{2}$ uniformly on $\{|\zeta| = 1,\ |\arg\zeta| \le \frac{\pi}{3}\}$, proving the first claim. Writing $S_G(u) = \int_0^{\infty} \frac{2u}{t+u^2}\, \mathrm{d}n_G(t)$ with $n_G(t) := \#\{j : w_j \le t\}$, Karamata's Tauberian theorem yields $t^{-1/2} n_G(t) \to \frac{\tau_0}{\pi}$, so h_G has indicator $\frac{\tau_0}{2}|\cos\theta|$ [15, Ch. II, Thm. 2] and conjugate indicator diagram $[-\frac{\tau_0}{2}, \frac{\tau_0}{2}]$ [3, Thm. 5.4.1]; evenness of h_G gives $\mathcal{B}_G(-\zeta) = -\mathcal{B}_G(\zeta)$. By Pólya's representation $h_G(2u) = \frac{1}{2\pi\mathrm{i}} \oint_{\Gamma_{\tau_0,\varepsilon}} \mathcal{B}_G(\zeta)\, \mathrm{e}^{2u\zeta}\, \mathrm{d}\zeta$ and $\xi_{\mathrm{DH}}(s(z)+\zeta) = \int_{-\infty}^{\infty} \Phi_{\mathrm{DH}}(u)\, \mathrm{e}^{2u\zeta + \mathrm{i}zu}\, \mathrm{d}u$, compactness of $\Gamma_{\tau_0,\varepsilon}$ and the double-exponential decay of $\Phi_{\mathrm{DH}}(u)$ allow Fubini's theorem to exchange the two integrals, establishing (2.4).

(ii) Set $\sigma_z := \mathrm{Re}\, s(z) = \frac{1}{2} + \delta \in [-\frac{3}{2}, \frac{5}{2}]$. Since $\Omega(z) = \frac{1}{2}\log\frac{5\gamma}{2\pi} + \frac{\mathrm{i}\pi}{4} + \mathcal{O}(\gamma^{-1})$ lies inside $|\arg w| \le \frac{\pi}{4}$ for $\gamma \ge \gamma_0$, integrating $\frac{h_G'(w)}{h_G(w)} = \frac{\tau_0}{2} + o(1)$ gives $|h_G(\Omega(z))| \ge \gamma^{\frac{\tau_0}{4} - \frac{\eta}{16}}$ for $\gamma \ge \gamma_1(G,\eta)$. On $\zeta \in \Gamma_+$, we have $\mathrm{Re}(s(z)+\zeta) \ge \sigma_z + \frac{\tau_0}{2} - \eta \ge 1 + 2\eta$. Set $N_\gamma := \lfloor \gamma^{1/4} \rfloor$ and split $\mathcal{D}_{\mathrm{DH}}(s(z)+\zeta) = \sum_{n=1}^{N_\gamma} a_{\mathrm{DH}}(n) n^{-s(z)-\zeta} + R_{\mathrm{Dir}}(z,\zeta)$. Because $|R_{\mathrm{Dir}}(z,\zeta)| \le \sum_{n>N_\gamma} n^{-\sigma_z - \mathrm{Re}\,\zeta} \ll_\eta \gamma^{\frac{1}{4}(1-\sigma_z-\mathrm{Re}\,\zeta)}$ and $|\Gamma_{\mathrm{DH}}(s(z)+\zeta)/\Gamma_{\mathrm{DH}}(s(z))| \ll_G \gamma^{\frac{1}{2}\mathrm{Re}\,\zeta}$, on $\zeta \in \Gamma_+$ ($\mathrm{Re}\,\zeta \le \frac{\tau_0}{2} + \varepsilon$) the product exponent is $\frac{1}{4}\mathrm{Re}\,\zeta + \frac{1}{4}(1-\sigma_z) \le \frac{\tau_0}{4} + \frac{1}{4}(1 + \frac{\eta}{16} - \sigma_z - \frac{\tau_0}{2}) < \frac{\tau_0}{4} - \frac{5}{8}\eta$, so the R_{Dir} integral contributes $\mathcal{O}_{G,\eta}(\gamma^{-\eta/2})|\Gamma_{\mathrm{DH}}(s(z))\, h_G(\Omega(z))|$.

For $1 \le n \le N_\gamma$, Stirling's expansion at $\frac{s(z)+1}{2}$ gives $\Gamma_{\mathrm{DH}}(s(z)+\zeta)/\Gamma_{\mathrm{DH}}(s(z)) = \mathrm{e}^{\Omega(z)\zeta}(1 + R_{\mathrm{St}}(z,\zeta))$ with $|R_{\mathrm{St}}(z,\zeta)| \ll_G \gamma^{-1}$ uniformly on $\Gamma_{\tau_0,\varepsilon}$. Summing $|n^{-s(z)-\zeta}\mathrm{e}^{\Omega(z)\zeta} R_{\mathrm{St}}| \ll_G \gamma^{-1+\frac{\tau_0}{4}+\frac{\varepsilon}{2}} n^{-(1+2\eta)}$ over $1 \le n \le N_\gamma$ contributes $\mathcal{O}_{G,\eta}(\gamma^{-1/2})|h_G(\Omega(z))|$. Next, for $1 \le n \le N_\gamma \le \gamma^{1/4}$ and $\zeta \in \Gamma_{\tau_0,\varepsilon} \setminus \Gamma_+$ ($\mathrm{Re}\,\zeta - \frac{\tau_0}{2} \le -\eta$, $|\mathrm{Im}\,\zeta| \le \varepsilon$), we have $\mathrm{Re}(\Omega(z) - \log n) \ge \frac{1}{4}\log\gamma - C_1 > 0$, whence

$$\big|n^{-s(z)}\mathrm{e}^{(\Omega(z)-\log n)\zeta}\big| \ll n^{-(\sigma_z+\tau_0/2)}\, \mathrm{e}^{(\mathrm{Re}\,\Omega(z) - \log n)\frac{\tau_0}{2}}\, \mathrm{e}^{-\eta(\frac{1}{4}\log\gamma - C_1)} \ll_\eta n^{-(1+3\eta)}\, \gamma^{\frac{\tau_0}{4} - \frac{\eta}{4}}.$$

Re-attaching $\Gamma_{\tau_0,\varepsilon}\setminus\Gamma_+$ therefore incurs an error $\mathcal{O}_{G,\eta}(\gamma^{-3\eta/16})|h_G(\Omega(z))|$ and evaluates each contour integral to $\frac{1}{2\pi\mathrm{i}}\oint_{\Gamma_{\tau_0,\varepsilon}}\mathcal{B}_G(\zeta)\,\mathrm{e}^{(\Omega(z)-\log n)\zeta}\,\mathrm{d}\zeta = h_G(\Omega(z)-\log n)$, yielding

$$\frac{I_+(z)}{\Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))} = \sum_{n=1}^{N_\gamma} a_{\mathrm{DH}}(n)\,n^{-s(z)}\,\frac{h_G(\Omega(z)-\log n)}{h_G(\Omega(z))} + \mathcal{O}_{G,\eta}(\gamma^{-3\eta/16}). \tag{2.6}$$

(Note that for $G_t(w)=\mathrm{e}^{tw}$, $h_{G_t}(w)=\mathrm{e}^{tw^2/4}$ gives $h_{G_t}(\Omega-\log n)/h_{G_t}(\Omega) = n^{-\frac{t}{2}\Omega}\mathrm{e}^{\frac{t}{4}\log^2 n}$, recovering Dobner's modified Dirichlet series [9].) Finally, let $\delta_G(\gamma) := \sup\{|\frac{h_G'(w)}{h_G(w)} - \frac{\tau_0}{2}| : \operatorname{Re} w \ge \frac14\log\gamma - C_1,\ |\arg w| \le \frac{\pi}{3}\}$; by (i), $\lim_{\gamma\to\infty}\delta_G(\gamma)=0$. For $1\le n\le N_\gamma$ and γ large enough that $\delta_G(\gamma)\le\eta$, integrating along $[\Omega(z)-\log n,\Omega(z)]$ gives $h_G(\Omega(z)-\log n)/h_G(\Omega(z)) = n^{-\tau_0/2+\theta_n(z)}$ with $|\theta_n(z)|\le\delta_G(\gamma)$. Given any $\varepsilon_1>0$, choosing M_1 so $\sum_{n>M_1} n^{-(1+2\eta)} < \varepsilon_1/3$ bounds both tails $n>M_1$ uniformly by $\varepsilon_1/3$, while on $1\le n\le M_1$ we have $\sum_{n=1}^{M_1}|n^{\theta_n(z)}-1|<\varepsilon_1/3$ once $\delta_G(\gamma)$ is sufficiently small, proving (2.5). □

Theorem 2.6 (Sufficient Stieltjes shift conditions on $\mathcal{G}_0$)**.** *Under the hypotheses of Theorem 1.1, assertions* (i)(a), (i)(b)*, and the positive half of* (i)(c) *hold for every $G\in\mathcal{G}_0$.*

Proof. In (a), $\frac{2u}{w_j+u^2}\le w_j^{-1/2}$ for all $u>0$, and $\sum_{j=1}^\infty w_j^{-1/2}<\infty$ gives $\lim_{u\to\infty}S_G(u)=0$ by dominated convergence. In (c), if $\lambda>0$ then $S_G(u)\ge 4\lambda u\to+\infty$; if $\lambda=0$ and $\underline{\rho}(G)>\frac12$, fix $\alpha\in(\frac12,\underline{\rho}(G))$. Since $u\mapsto u\,S_G(u)=\sum_{j=1}^\infty\frac{2u^2}{w_j+u^2}$ is strictly increasing on $(0,\infty)$,

$$\log G(u^2)-\log G(1) = \int_1^u \frac{y\,S_G(y)}{y}\,\mathrm{d}y \le u\,S_G(u)\log u \implies S_G(u)\ge\frac{u^{2\alpha}-\mathcal{O}(1)}{u\log u}\longrightarrow+\infty \tag{2.7}$$

as $u\to+\infty$, because $2\alpha>1$.

Assume now that $\lambda=0$ and $\lim_{u\to\infty}S_G(u)=\tau_0\in[0,\infty)$, covering both (a) ($\tau_0=0$) and (b) ($\tau_0=\tau>0$).

Case 1: Subcritical persistence ($0\le\tau_0<2\sigma^*-1$). Set $\sigma_{\mathrm{low}} := \max(1,\frac12+\frac{\tau_0}{2})<\sigma^*$. Applying Lemma 2.4(ii) provides a zero $\rho_*=\beta_0+\mathrm{i}\gamma_*$ with $\beta_0\in(\sigma_{\mathrm{low}},\sigma^*]$, $\eta_0=\frac14(\beta_0-\sigma_{\mathrm{low}})\in(0,\frac{1}{16})$ (shrinking η_0 if necessary), $r_0\in(0,\min(\eta_0/2,1/4))$, $\mu_0>0$, and $M(T)\gg_G T$ unit-separated shifts $\tau_m\in[\frac12T,\frac34T]$ (T large) satisfying (2.3) on $|w-s_m|=r_0$, where $s_m:=\rho_*+\mathrm{i}\tau_m=\beta_0+\mathrm{i}\gamma_m$ ($\gamma_m:=\gamma_*+\tau_m\in[\frac13T,T]$). Set $z_m^{\mathrm{sh}}:=2\gamma_m-2\mathrm{i}(\beta_0-\frac12-\frac{\tau_0}{2})$, so $s(z_m^{\mathrm{sh}})+\frac{\tau_0}{2}=s_m$ and $|s(z)+\frac{\tau_0}{2}-s_m|=\frac12|z-z_m^{\mathrm{sh}}|$. The disks $|z-z_m^{\mathrm{sh}}|\le 2r_0$ are pairwise disjoint and satisfy $\operatorname{Im} z\le -2(\beta_0-\frac12-\frac{\tau_0}{2})+2r_0\le-7\eta_0<0$ (and $\operatorname{Im} z<-1$ when $\tau_0=0$, since $\beta_0-r_0>1$). Moreover, on $|z-z_m^{\mathrm{sh}}|\le 2r_0$ we have $\sigma_z:=\operatorname{Re} s(z)\ge\beta_0-\frac{\tau_0}{2}-r_0>\frac12+\frac72\eta_0$ and $\sigma_z+\frac{\tau_0}{2}>1+\frac72\eta_0$. Applying Lemma 2.5 with $\eta=\eta_0$ and $\varepsilon=\frac{\eta_0}{16}$ gives (2.5) on Γ_+. On $\zeta\in\Gamma_{\tau_0,\varepsilon}\setminus\Gamma_+$ ($-\frac{\tau_0}{2}-\varepsilon\le\operatorname{Re}\zeta\le\frac{\tau_0}{2}-\eta_0$), Lemma 2.4(i) with parameter $\frac{\eta_0}{2}$ bounds $|\xi_{\mathrm{DH}}(s(z)+\zeta)|\ll_{G,\eta_0}\gamma_m^{\frac12\max(\sigma,1-\sigma,1+\eta_0/2)}\mathrm{e}^{-\frac{\pi}{4}\gamma_m}$, where $\sigma=\sigma_z+\operatorname{Re}\zeta$ satisfies

$$\Big(\frac12\sigma_z+\frac{\tau_0}{4}\Big)-\frac12\max\Big(\sigma,\,1-\sigma,\,1+\frac{\eta_0}{2}\Big) = \min\Big(\tfrac12(\tfrac{\tau_0}{2}-\operatorname{Re}\zeta),\ \sigma_z-\tfrac12+\tfrac12(\tfrac{\tau_0}{2}+\operatorname{Re}\zeta),\ \tfrac12(\sigma_z+\tfrac{\tau_0}{2}-1-\tfrac{\eta_0}{2})\Big)\ge\frac{\eta_0}{2},$$

so $\Gamma_{\tau_0,\varepsilon}\setminus\Gamma_+$ contributes $\mathcal{O}_{G,\eta_0}(\gamma_m^{-7\eta_0/16})|\Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))|$. Hence uniformly on $|z-z_m^{\mathrm{sh}}|=2r_0$, $F_G(z)/[\Gamma_{\mathrm{DH}}(s(z))\,h_G(\Omega(z))] = \mathcal{D}_{\mathrm{DH}}(s(z)+\frac{\tau_0}{2})+o_G(1)$, whereas $|\mathcal{D}_{\mathrm{DH}}(s(z)+\frac{\tau_0}{2})|>\frac{\mu_0}{2}$ by (2.3). For T sufficiently large, Rouché's theorem yields at least one zero of F_G inside each disk $|z-z_m^{\mathrm{sh}}|<2r_0\subset\{\operatorname{Im} z<0\}$, proving $N_{\mathrm{off}}(F_G;T)\gg_G T$.

Case 2: Supercritical finiteness ($\tau_0=\tau>2\sigma^*-1$). By $F_G(-z)=F_G(z)$, $F_G(\bar z)=\overline{F_G(z)}$, and Lemma 2.1(i), every zero satisfies $|\delta|\le h_\infty^*$ ($z=x-2\mathrm{i}\delta$, $x=2\gamma$), and it suffices to find $X_0>0$ such that all zeros of $F_G(z)$ in $\operatorname{Re} z\ge X_0$ are real. Choose $\eta_2\in(0,\min(\frac{1}{16},\frac16(\frac12+\frac{\tau}{2}-\sigma^*)))$ and $\varepsilon:=\frac{\eta_2}{16}$. For all $\delta\in[-2\eta_2,h_\infty^*]$ and $\gamma\ge\gamma_1$, we have $\operatorname{Re}(s(z)+\frac{\tau}{2})=\frac12+\delta+\frac{\tau}{2}\ge$

$\sigma^* + 4\eta_2 > 1 + 4\eta_2$, so Lemma 2.4(iii) gives $|\mathcal{D}_{\mathrm{DH}}(s(z) + \frac{\tau}{2})| \geq c(4\eta_2) > 0$. Consequently,

$$\mathcal{M}_G(z) := \Gamma_{\mathrm{DH}}(s(z))\, h_G(\Omega(z))\, \mathcal{D}_{\mathrm{DH}}\big(s(z) + \tfrac{\tau}{2}\big) \tag{2.8}$$

is holomorphic and non-vanishing on $\{\gamma \geq \gamma_1,\ \delta \in [-2\eta_2, h^*_\infty]\}$, and Lemma 2.5(ii) gives $I_+(z) = \mathcal{M}_G(z)(1 + o(1))$ uniformly there as $\gamma \to +\infty$.

First, on the outer strip $\delta \in [\frac{\eta_2}{4}, h^*_\infty]$ ($\sigma_z - \frac{1}{2} = \delta \geq \frac{\eta_2}{4}$), Lemma 2.4(i) with parameter $\frac{\eta_2}{2}$ bounds $\Gamma_{\tau,\varepsilon} \setminus \Gamma_+$ by $\mathcal{O}_{G,\eta_2}(\gamma^{-5\eta_2/32})|\mathcal{M}_G(z)|$ (since $\delta + \frac{1}{2}(\frac{\tau}{2} + \operatorname{Re}\zeta) \geq \frac{\eta_2}{4} - \frac{\eta_2}{32} = \frac{7}{32}\eta_2$ and $|h_G(\Omega(z))| \geq \gamma^{\frac{\tau}{4} - \frac{\eta_2}{16}}$), so $F_G(z) = \mathcal{M}_G(z)(1 + o(1)) \neq 0$ for $\gamma \geq \gamma_2$, and by conjugation $F_G(z) \neq 0$ on $[-h^*_\infty, -\frac{\eta_2}{4}]$.

Second, on the remaining strip $|\delta| \leq \frac{\eta_2}{4}$, split $\Gamma_{\tau,\varepsilon} = \Gamma_+ \cup \Gamma_- \cup \Gamma_{\mathrm{mid}}$, where $\Gamma_- := -\Gamma_+$ ($\operatorname{Re}\zeta \leq -\frac{\tau}{2} + \eta_2$) and $\Gamma_{\mathrm{mid}} := \{\zeta \in \Gamma_{\tau,\varepsilon} : |\operatorname{Re}\zeta| < \frac{\tau}{2} - \eta_2\}$. Because $\mathcal{B}_G(-\zeta) = -\mathcal{B}_G(\zeta)$, $\mathcal{B}_G(\bar{w}) = \overline{\mathcal{B}_G(w)}$, and $\xi_{\mathrm{DH}}(s(z) - u) = \xi_{\mathrm{DH}}(1 - s(z) + u) = \overline{\xi_{\mathrm{DH}}(s(\bar{z}) + \bar{u})}$, substituting $\zeta = -\bar{w}$ yields the exact identity $I_-(z) = \overline{I_+(\bar{z})}$. Since both z and $\bar{z}$ have ordinate in $[-\frac{\eta_2}{4}, \frac{\eta_2}{4}] \subset [-2\eta_2, h^*_\infty]$, we obtain $I_+(z) + I_-(z) = \mathcal{M}_G(z)(1 + o(1)) + \overline{\mathcal{M}_G(\bar{z})}(1 + o(1))$, while Lemma 2.4(i) on Γ_{mid} gives $|I_{\mathrm{mid}}(z)| \ll_{G,\eta_2} \gamma^{-7\eta_2/16} \max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar{z})}|)$. Thus uniformly on $|\delta| \leq \frac{\eta_2}{4}$ as $\gamma \to +\infty$,

$$F_G(z) = H_{\mathrm{main}}(z) + E_G(z), \qquad H_{\mathrm{main}}(z) := \mathcal{M}_G(z) + \overline{\mathcal{M}_G(\bar{z})}, \tag{2.9}$$

where $|E_G(z)| = o_{G,\eta_2}(1) \cdot \max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar{z})}|)$. Differentiating $\log \mathcal{M}_G(z)$ on $|\delta| \leq \frac{\eta_2}{2}$ gives: $\frac{\mathrm{d}}{\mathrm{d}z} \log \Gamma_{\mathrm{DH}}(s(z)) = \frac{\mathrm{i}}{4}[\log \frac{5}{\pi} + \psi(\frac{s(z)+1}{2})] = -\frac{\pi}{8} + \frac{\mathrm{i}}{4} \log \frac{5\gamma}{2\pi} + \mathcal{O}(\gamma^{-1})$; $\frac{\mathrm{d}}{\mathrm{d}z} \log h_G(\Omega(z)) = \frac{\mathrm{i}}{4(s(z)+1)}(\frac{\tau}{2} + o(1)) = \mathcal{O}_G(\gamma^{-1})$; and $|\frac{\mathrm{d}}{\mathrm{d}z} \log \mathcal{D}_{\mathrm{DH}}(s(z) + \frac{\tau}{2})| \leq \frac{-\zeta'(1+4\eta_2)}{2c(4\eta_2)} = \mathcal{O}_{\eta_2}(1)$. Hence uniformly on $|\delta| \leq \frac{\eta_2}{2}$, $\gamma \geq \gamma_2$,

$$\frac{\mathrm{d}}{\mathrm{d}z} \log \mathcal{M}_G(z) = -\frac{\pi}{8} + \frac{\mathrm{i}}{4} \log \frac{5\gamma}{2\pi} + \mathcal{O}_{G,\eta_2}(1). \tag{2.10}$$

For fixed $x = 2\gamma$, $\frac{\partial}{\partial u} \log |\mathcal{M}_G(x - 2\mathrm{i}u)| = 2 \operatorname{Im}(\frac{\mathcal{M}'_G}{\mathcal{M}_G}(x - 2\mathrm{i}u)) = \frac{1}{2} \log \frac{5\gamma}{2\pi} + \mathcal{O}_{G,\eta_2}(1) \geq \frac{3}{8} \log \gamma$ for $\gamma \geq \gamma_3$, so integrating over $[-\delta, \delta]$ yields $\log |\mathcal{M}_G(x - 2\mathrm{i}\delta)/\overline{\mathcal{M}_G(x + 2\mathrm{i}\delta)}| \geq \frac{3}{4}\delta \log \gamma > 0$ for every $\delta \in (0, \frac{\eta_2}{4}]$. This has two consequences for $x = 2\gamma \geq X_0$ (X_0 chosen so $|E_G(z)| < \frac{1}{4} \max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar{z})}|)$ on $|\delta| \leq \frac{\eta_2}{4}$):

(a) $H_{\mathrm{main}}(z)$ has no non-real zeros in $\{x \geq X_0,\ |\delta| \leq \frac{\eta_2}{4}\}$, and whenever $\frac{4}{\log(x/2)} \leq |\delta| \leq \frac{\eta_2}{4}$, we have $|H_{\mathrm{main}}(z)| \geq (1 - \mathrm{e}^{-3}) \max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar{z})}|) > \frac{9}{10} \max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar{z})}|) > |E_G(z)|$, excluding zeros of $F_G(z)$ in $\frac{4}{\log(x/2)} \leq |\delta| \leq \frac{\eta_2}{4}$.

(b) On $\mathbb{R} \cap [X_0, \infty)$, writing $\mathcal{M}_G(x) = |\mathcal{M}_G(x)|\mathrm{e}^{\mathrm{i}\vartheta(x)}$ gives $\vartheta'(x) = \operatorname{Im}(\frac{\mathcal{M}'_G(x)}{\mathcal{M}_G(x)}) = \frac{1}{4} \log \frac{5x}{4\pi} + \mathcal{O}_{G,\eta_2}(1) > \frac{1}{8} \log x > 0$, so the points x_k ($k \geq k_0$) with $\vartheta(x_k) = k\pi$ satisfy $x_{k+1} - x_k \asymp \frac{4\pi}{\log x_k}$ and separate the simple real zeros $r_k \in (x_k, x_{k+1})$ of $H_{\mathrm{main}}(x) = 2|\mathcal{M}_G(x)| \cos \vartheta(x)$. Set $\delta_k := \frac{4}{\log(x_k/2)}$ and $\mathcal{R}_k := [x_k, x_{k+1}] \times [-2\mathrm{i}\delta_k, 2\mathrm{i}\delta_k]$. Because $\delta_k \geq \frac{4}{\log(x/2)}$ for all $x \in [x_k, x_{k+1}]$, the union $\bigcup_{k \geq k_0} \mathcal{R}_k$ covers $\{x \geq x_{k_0},\ |\delta| \leq \frac{4}{\log(x/2)}\}$ with no corner gap, and (a) already gives $|H_{\mathrm{main}}(z)| > |E_G(z)|$ along both horizontal edges $z = x \mp 2\mathrm{i}\delta_k$. Along the vertical edges $z = x_j - 2\mathrm{i}\delta$ ($j \in \{k, k+1\}$, $|\delta| \leq \delta_k$), integrating (2.10) from x_j gives $\mathcal{M}_G(x_j - 2\mathrm{i}\delta) = (-1)^j |\mathcal{M}_G(x_j - 2\mathrm{i}\delta)| \mathrm{e}^{\mathrm{i}\frac{\pi}{4}\delta + \mathcal{O}(1/\log x_k)}$ and $\overline{\mathcal{M}_G(x_j + 2\mathrm{i}\delta)} = (-1)^j |\mathcal{M}_G(x_j + 2\mathrm{i}\delta)| \mathrm{e}^{\mathrm{i}\frac{\pi}{4}\delta + \mathcal{O}(1/\log x_k)}$; since the two summands form an angle $\mathcal{O}(1/\log x_k) < \frac{\pi}{6}$, we have $|H_{\mathrm{main}}(z)| \geq \max(|\mathcal{M}_G(z)|, |\overline{\mathcal{M}_G(\bar{z})}|) > |E_G(z)|$ on both vertical edges as well.

By Rouché's theorem, F_G has a unique zero (counted with multiplicity) inside $\operatorname{int}(\mathcal{R}_k)$ for every $k \geq k_0$; by conjugation symmetry $F_G(\bar{z}) = \overline{F_G(z)}$ of $\mathcal{R}_k$, this unique zero must be real and simple (confirming also via Cauchy's estimate $|E'_G(x)| = o(\log x)|\mathcal{M}_G(x)|$ on disks of radius $\frac{4}{\log(x/2)}$ that F'_G does not vanish there). Thus $N_{\mathrm{off}}(F_G; \infty) < \infty$.

Finally, in (c), if $\lambda > 0$, write $G(w) = \mathrm{e}^{\lambda w} G_1(w)$ with $G_1 \in \mathcal{G}_0$; Proposition 4.3 gives $N_{\mathrm{off}}(H^{\mathrm{DH}}_\lambda; \infty) < \infty$, and Lemma 2.2 yields $N_{\mathrm{off}}(G(-\partial_z^2)\Xi_{\mathrm{DH}}; \infty) \leq N_{\mathrm{off}}(H^{\mathrm{DH}}_\lambda; \infty) < \infty$.

If $\lambda = 0$ and $w_{j+1} - w_j = o(\sqrt{w_j})$, set $\tau_1 := 2\sigma^* > 2\sigma^* - 1$ and $t_m := (\frac{\pi m}{\tau_1})^2$. Since $t_{m+1} - t_m = \frac{2\pi}{\tau_1}\sqrt{t_m} + \frac{\pi^2}{\tau_1^2}$ dominates $o(\sqrt{t_m})$, every open interval (t_m, t_{m+1}) $(m \geq m_0)$ contains at least one root w_{j_m}. Disjointness of (t_m, t_{m+1}) ensures $j_{m_0} < j_{m_0+1} < \cdots$ and $w_{j_m}/t_m \to 1$, so $G_{\mathrm{sub}}(w) := \prod_{m=m_0}^{\infty}(1 + w/w_{j_m})$ satisfies $\lim_{u\to\infty} S_{G_{\mathrm{sub}}}(u) = \tau_1 > 2\sigma^* - 1$, giving $N_{\mathrm{off}}(G_{\mathrm{sub}}(-\partial_z^2)\Xi_{\mathrm{DH}}; \infty) < \infty$, and Lemma 2.2 applied to $G/G_{\mathrm{sub}} \in \mathcal{G}_0$ completes the proof. □

Proposition 2.7 (Lacunary canonical product counterexample)**.** *Let $\rho \in (1/2, 1)$, define recursively $R_1 := 16$ and $R_k := 2^{2^{R_{k-1}}}$ for $k \geq 2$, and set $M_k := \lfloor R_k^{2\rho} \rfloor$. Then $G_{\mathrm{lac}}(w) := \prod_{k=1}^{\infty}(1 + w/R_k^2)^{M_k} \in \mathcal{G}_0$ has upper order $\rho(G_{\mathrm{lac}}) = \rho$ and lower order $\underline{\rho}(G_{\mathrm{lac}}) = 0$, satisfies $\lim_{k\to\infty} S_{G_{\mathrm{lac}}}(u_k) = 0$ along $u_k := R_k^{(1-\rho)/2}$, and $G_{\mathrm{lac}}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ possesses infinitely many non-real zeros inside $|\operatorname{Im} z| > 1$.*

Proof. Because $\sum_{k=1}^{\infty} M_k R_k^{-2} \leq \sum_{k=1}^{\infty} R_k^{2\rho-2} =: C_{\mathrm{lac}} < \infty$ and $n_{G_{\mathrm{lac}}}(t) \leq \sum_{R_j \leq \sqrt{t}} R_j^{2\rho} \ll t^{\rho}$ with $n_{G_{\mathrm{lac}}}(R_k^2) \geq M_k \sim (R_k^2)^{\rho}$, Borel's theorem [3, Thm. 2.6.5] gives $G_{\mathrm{lac}} \in \mathcal{G}_0$ and $\rho(G_{\mathrm{lac}}) = \rho$. Since $R_{k-1} = \log_2 \log_2 R_k$ $(k \geq 2)$, the prefix degree satisfies $D_{k-1} := \sum_{j=1}^{k-1} M_j \ll (\log\log R_k)^{2\rho}$. At $u_k = R_k^{(1-\rho)/2}$, $\log G_{\mathrm{lac}}(u_k^2) \leq D_{k-1}\log(1 + u_k^2) + \sum_{j=k}^{\infty} M_j u_k^2 R_j^{-2} \ll (\log\log R_k)^{2\rho}\log R_k + R_k^{\rho-1}$, whence $\underline{\rho}(G_{\mathrm{lac}}) = 0$, and $S_{G_{\mathrm{lac}}}(u_k) \leq \frac{2D_{k-1}}{u_k} + \sum_{j=k}^{\infty} \frac{2M_j u_k}{R_j^2} \ll (\log\log R_k)^{2\rho} R_k^{-(1-\rho)/2} + R_k^{-\frac{3}{2}(1-\rho)} \to 0$.

To produce non-real zeros in $|\operatorname{Im} z| > 1$, apply Lemma 2.4(ii) with $\sigma_{\mathrm{low}} = 1$ to obtain a fixed zero $\rho_* = \beta_0 + \mathrm{i}\gamma_*$ $(\beta_0 \in (1, \sigma^*])$, $\eta_0 = \frac{1}{4}(\beta_0 - 1)$, $r_0 \in (0, \min(\eta_0/2, 1/4))$, and $\mu_0 > 0$. For each large k, set $T_k := R_k^{(1-\rho)/4}$ and choose via Lemma 2.4(ii) a shift $\tau^{(k)} \in [T_k, 2T_k]$ so that $|\mathcal{D}_{\mathrm{DH}}(w)| > \frac{\mu_0}{2}$ on $|w - s_k| = r_0$ and $\mathcal{D}_{\mathrm{DH}}$ vanishes inside $|w - s_k| < r_0$, where $s_k := \rho_* + \mathrm{i}\tau^{(k)} = \beta_0 + \mathrm{i}\gamma_k$ $(\gamma_k \asymp T_k)$. Set $z_k := 2\gamma_k - 2\mathrm{i}(\beta_0 - \frac{1}{2})$, so $|z - z_k| \leq 2r_0$ lies inside $\operatorname{Im} z \leq -1 - 7\eta_0 < -1$, and factor $G_{\mathrm{lac}} = G_{\leq k-1}G_{\geq k}$. Shifting $u = v + \mathrm{i}\alpha_k$ with $\alpha_k := \frac{\pi}{8} - \frac{1}{\gamma_k}$ in $\Delta_k(z) := \int_{-\infty}^{\infty} G_{\leq k-1}(u^2)(G_{\geq k}(u^2) - 1)\Phi_{\mathrm{DH}}(u)\mathrm{e}^{\mathrm{i}zu}\,\mathrm{d}u$ (legitimate on $|\operatorname{Im} u| < \frac{\pi}{8}$ by the double-exponential decay of Φ_{DH}) extracts $|\mathrm{e}^{\mathrm{i}z(v+\mathrm{i}\alpha_k)}| \asymp \mathrm{e}^{-\frac{\pi}{4}\gamma_k}\mathrm{e}^{2\delta v}$. Using $|\Phi_{\mathrm{DH}}(v + \mathrm{i}\alpha_k)| \ll \gamma_k \mathrm{e}^{-|v|}$ on $|v| \leq V_k := \log\gamma_k$ and $\ll \mathrm{e}^{3|v| - \frac{2}{5}\gamma_k^{-1}\mathrm{e}^{4|v|}}$ on $|v| > V_k$, together with $|G_{\leq k-1}(u^2)| \leq (2 + V_k^2)^{D_{k-1}} = R_k^{o(1)}$ and $|G_{\geq k}(u^2) - 1| \ll (\log R_k)^2 R_k^{2\rho-2}$ on $|v| \leq V_k$, dividing by $|\Gamma_{\mathrm{DH}}(s(z))| \geq \gamma_k^{1/2}\mathrm{e}^{-\frac{\pi}{4}\gamma_k}$ yields $|\Delta_k(z)/\Gamma_{\mathrm{DH}}(s(z))| \ll R_k^{2\rho-2+\frac{1-\rho}{4}(2\sigma^*-1/2)+o(1)} \leq R_k^{-\frac{3}{2}(1-\rho)+o(1)} \to 0$ (as $2\sigma^* - \frac{1}{2} < 2$). Meanwhile, for the polynomial $G_{\leq k-1}$ of degree $D_{k-1} \ll (\log\log\gamma_k)^{2\rho}$, writing $\xi_{\mathrm{DH}}(s + \zeta)/\Gamma_{\mathrm{DH}}(s) = \mathrm{e}^{\Omega(z)\zeta} f_s(\zeta)$ and applying Cauchy's bounds $|f_s^{(j)}(0)|/\Gamma(j + 1) \leq C_2(\eta_0/4)^{-j}$ on $|\zeta| \leq \eta_0/4$ alongside $G_{\leq k-1}(\frac{1}{4}|\Omega(z)|^2) \leq \mathrm{e}^{\frac{1}{2}C_{\mathrm{lac}}}|G_{\leq k-1}(\frac{1}{4}\Omega(z)^2)|$ and $|G_{\leq k-1}(\frac{1}{4}\Omega(z)^2)| \geq 1$ gives $G_{\leq k-1}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)/[\Gamma_{\mathrm{DH}}(s(z))G_{\leq k-1}(\frac{1}{4}\Omega(z)^2)] = \mathcal{D}_{\mathrm{DH}}(s(z)) + \mathcal{O}(D_{k-1}/\log\gamma_k)$ on $|z - z_k| = 2r_0$. Rouché's theorem produces a zero of $G_{\mathrm{lac}}(-\partial_z^2)\Xi_{\mathrm{DH}}$ in $|z - z_k| < 2r_0 \subset \{\operatorname{Im} z < -1\}$ for every large k. □

Remark 2.8 (Open boundary regimes). Theorem 2.6 and Proposition 2.7 provide sufficient conditions rather than an exhaustive characterization of $\mathcal{G}_0$. Three boundary regimes remain open:

(a) multipliers with $\lambda = 0$ whose Stieltjes derivative oscillates across the threshold, $\liminf_{u\to\infty} S_G(u) < 2\sigma^* - 1 < \limsup_{u\to\infty} S_G(u)$, without root-gap regularity;
(b) the critical boundary case $\lambda = 0$ and $\lim_{u\to\infty} S_G(u) = 2\sigma^* - 1$, where the shifted abscissa $\frac{1}{2} + \frac{\tau_0}{2} = \sigma^*$ meets the supremum of off-line zeros without a uniform positive lower bound on $|\mathcal{D}_{\mathrm{DH}}|$; and
(c) multipliers with $\lambda = 0$ and lower order $\underline{\rho}(G) > \frac{1}{2}$ (so $\lim_{u\to\infty} S_G(u) = +\infty$) whose roots contain clusters violating $w_{j+1} - w_j = o(\sqrt{w_j})$ and admitting no supercritical regular subproduct.

3. Global versus Asymptotic Threshold Separation for Pólya's Sinc Multiplier

Both $G_{\cosh,h}(w) = \cosh(2h\sqrt{w})$ and Pólya's hyperbolic sinc multiplier $G_{\mathrm{sinc},h}(w) = \frac{\sinh(2h\sqrt{w})}{2h\sqrt{w}} = \prod_{j=1}^{\infty}(1+\frac{4h^2w}{\pi^2 j^2})$ belong to $\mathcal{G}_0$ with identical asymptotic shift $\lim_{u\to\infty} S_G(u) = 2h$ and identical infinite-height threshold $h^*_\infty = \sigma^* - \frac{1}{2}$. In this section we prove Theorem 1.1(ii) in a manner completely independent of Theorem 2.6.

Lemma 3.1 (Enclosure of σ^*, strict zero inequality, and global threshold of $G_{\cosh,h}$)**.** *The unique root $\sigma^* > 1$ of the Bombieri–Ghosh prime-angle identity $\sum_{p\equiv 2,3 \pmod 5} \arctan(p^{-\sigma^*}) = \frac{\pi}{2} - \arctan\kappa$ satisfies*

$$\sigma^* \in \big[1.120362498183325061801458922264 64 \pm 10^{-30}\big]. \tag{3.1}$$

Moreover, every zero $\rho = \sigma + \mathrm{i}\gamma$ of $\mathcal{D}_{\mathrm{DH}}(s)$ satisfies $\sigma < \sigma^*$ strictly, every zero of $\Xi_{\mathrm{DH}}(z)$ satisfies $|\operatorname{Im} z| < 2(\sigma^* - \frac{1}{2})$, and

$$h^*_{\mathrm{global}}(G_{\cosh}; \Xi_{\mathrm{DH}}) = \sigma^* - \tfrac{1}{2} \in \Big(\frac{310}{500}, \frac{311}{500}\Big). \tag{3.2}$$

Proof. Summing $p < 1000$ directly and expanding $\arctan(p^{-\sigma}) = \sum_{m=0}^{14} \frac{(-1)^m}{2m+1} p^{-(2m+1)\sigma} + \mathcal{O}(10^{-95})$ for $p \geq 1000$ expresses each odd prime-power sum via Möbius inversion of $\frac{1}{2}\log[(1-5^{-u})\zeta(u)/L(u,\chi_{5,\mathrm{quad}})]$, where $\chi_{5,\mathrm{quad}}$ is the real non-principal character modulo 5. Evaluating at $1.120362498183325061801458922264 64 \pm 5\times 10^{-31}$ in 320-bit `FLINT/Arb` arithmetic certifies opposite signs by the Intermediate Value Theorem, proving (3.1).

Next, let $\rho = \sigma + \mathrm{i}\gamma$ be any zero of $\mathcal{D}_{\mathrm{DH}}(s)$. If $\sigma \leq 1$, then $\sigma < \sigma^*$. If $\sigma > 1$, then $L(s,\chi_5)$ and $L(s,\overline{\chi_5})$ do not vanish, and $\mathcal{D}_{\mathrm{DH}}(\sigma+\mathrm{i}\gamma) = 0$ is equivalent to

$$\frac{L(\sigma+\mathrm{i}\gamma, \overline{\chi_5})}{L(\sigma+\mathrm{i}\gamma, \chi_5)} = -\frac{1-\mathrm{i}\kappa}{1+\mathrm{i}\kappa} = \mathrm{e}^{+2\mathrm{i}(\frac{\pi}{2}-\arctan\kappa)}. \tag{3.3}$$

In the Euler product for (3.3), primes $p \equiv 1,4 \pmod 5$ have $\overline{\chi_5(p)} = \chi_5(p)$ and cancel identically, while each $p \equiv 2,3 \pmod 5$ contributes $\frac{1-w_p}{1+w_p}$ with $w_p := \chi_5(p)p^{-\sigma-\mathrm{i}\gamma}$ ($|w_p| = p^{-\sigma} < 1$). The principal argument satisfies $|\arg\frac{1-w_p}{1+w_p}| \leq 2\arctan(p^{-\sigma})$, with equality $\arg\frac{1-w_p}{1+w_p} = +2\arctan(p^{-\sigma})$ if and only if $w_p = -\mathrm{i}p^{-\sigma}$. Since $\sum_{p\equiv 2,3 \pmod 5} 2\arctan(p^{-\sigma}) \leq 2(\frac{\pi}{2} - \arctan\kappa) < \pi$ whenever $\sigma \geq \sigma^*$, equality in (3.3) at $\sigma \geq \sigma^*$ requires $\sigma = \sigma^*$ and $w_p = -\mathrm{i}p^{-\sigma^*}$ for every $p \equiv 2,3 \pmod 5$, in particular for both $p = 2$ and $p = 3$. When $\gamma = 0$, $w_2 = +\mathrm{i}2^{-\sigma} \neq -\mathrm{i}2^{-\sigma}$ gives $\arg\frac{1-w_2}{1+w_2} = -2\arctan(2^{-\sigma}) < 0$ (opposite in sign to $\arg\frac{1-w_3}{1+w_3} = +2\arctan(3^{-\sigma}) > 0$), so equality fails at $\gamma = 0$. When $\gamma \neq 0$, simultaneous equality $w_2 = \mathrm{i}2^{-\sigma-\mathrm{i}\gamma} = -\mathrm{i}2^{-\sigma}$ and $w_3 = -\mathrm{i}3^{-\sigma-\mathrm{i}\gamma} = -\mathrm{i}3^{-\sigma}$ requires $2^{-\mathrm{i}\gamma} = -1$ and $3^{-\mathrm{i}\gamma} = +1$, so $\gamma\log 2 = (2k_1+1)\pi$ and $\gamma\log 3 = 2k_2\pi$ ($k_1 \in \mathbb{Z}$, $k_2 \in \mathbb{Z}\setminus\{0\}$), contradicting $\frac{\log 3}{\log 2} \notin \mathbb{Q}$. Thus $\sigma < \sigma^*$ strictly; by $\xi_{\mathrm{DH}}(s) = \xi_{\mathrm{DH}}(1-s)$ and $\sigma^* > 1$, every zero of $\Xi_{\mathrm{DH}}(z)$ satisfies $|\operatorname{Im} z| < 2(\sigma^* - \frac{1}{2})$.

Consequently, Lemma 2.1(ii) implies that $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ has only real zeros for every $h \geq \sigma^* - \frac{1}{2}$, so $h^*_{\mathrm{global}}(G_{\cosh};\Xi_{\mathrm{DH}}) \leq \sigma^* - \frac{1}{2}$. Conversely, for any $0 < h < \sigma^* - \frac{1}{2}$, write $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \frac{1}{2}[\xi_{\mathrm{DH}}(s(z)+h) + \xi_{\mathrm{DH}}(s(z)-h)]$ and apply Lemma 2.4(ii) with $\sigma_{\mathrm{low}} := \max(1, \frac{1}{2}+h) < \sigma^*$ to obtain shifted disks $|s(z) + h - (\rho_* + \mathrm{i}\tau_m)| \leq r_0$ inside $\operatorname{Im} z < 0$ on whose boundary $|\mathcal{D}_{\mathrm{DH}}(s(z)+h)| > \frac{\mu_0}{2}$. By Lemma 2.4(i), $|\xi_{\mathrm{DH}}(s(z)-h)/\Gamma_{\mathrm{DH}}(s(z)+h)| \ll_h \gamma_m^{-\eta_0} \to 0$ as $\tau_m \to +\infty$, so Rouché's theorem produces non-real zeros of $G_{\cosh,h}(-\partial_z^2)\Xi_{\mathrm{DH}}$ for every $0 < h < \sigma^* - \frac{1}{2}$, proving (3.2). □

Theorem 3.2 (Global bounds and strict threshold separation for $G_{\mathrm{sinc},h}$)**.**

(i) *For every $h \in [0, \frac{311}{500}]$, the entire function $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ possesses at least one non-real zero; at $h = \frac{311}{500} = 0.622 > \sigma^* - \frac{1}{2}$, it possesses a certified simple non-real zero*

in the box $B_{\mathrm{sinc}} := \{z \in \mathbb{C} : |\operatorname{Re}(z - z_{\mathrm{sinc}})| \le 10^{-30},\ |\operatorname{Im}(z - z_{\mathrm{sinc}})| \le 10^{-30}\}$ *centered at*

$$\begin{aligned} z_{\mathrm{sinc}} = {} & 4884.886654191757156433333001976151 78 \\ & - 0.056004628860858072317639148896287841\,\mathrm{i}, \end{aligned} \tag{3.4}$$

originating at $h = 0$ *from the zero* $\rho_{2442} = 0.89775015\ldots + 2442.53029247\ldots\,\mathrm{i}$ *of* $\mathcal{D}_{\mathrm{DH}}(s)$.

(ii) *Conversely, for every* $h \ge 31$*, all zeros of* $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z)$ *are real.*

Consequently, $\sigma^* - \frac{1}{2} = h^*_{\mathrm{global}}(G_{\cosh}; \Xi_{\mathrm{DH}}) < \frac{311}{500} < h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \le 31$.

Proof. (i) Set $s_{\mathrm{ref}} := 0.71 + 2442.49\,\mathrm{i}$, $C_{\mathrm{ref}} := \Gamma_{\mathrm{DH}}(s_{\mathrm{ref}}) \ne 0$, and $\tilde{\xi}_{\mathrm{DH}}(s) := \xi_{\mathrm{DH}}(s)/C_{\mathrm{ref}}$. By Fubini's theorem on $G_{\mathrm{sinc},h}(u^2) = \frac{1}{2}\int_{-1}^{1} \mathrm{e}^{2huv}\,\mathrm{d}v$, we have for all $z \in \mathbb{C}$ and all $h \in \mathbb{R}$ (including $h = 0$, with no division by h) the singularity-free power-series identity

$$g_{\mathrm{sinc}}(z,h) := C_{\mathrm{ref}}^{-1}\, G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(z) = \frac{1}{2}\int_{-1}^{1} \tilde{\xi}_{\mathrm{DH}}(s(z) + hv)\,\mathrm{d}v = \sum_{m=0}^{\infty} \frac{\tilde{\xi}_{\mathrm{DH}}^{(2m)}(s(z))}{\Gamma(2m+2)}\, h^{2m}. \tag{3.5}$$

Covering the circle $|w - s_{\mathrm{ref}}| = \frac{9}{2}$ by 512 overlapping complex interval arcs in 320-bit `FLINT/Arb` arithmetic certifies $M_{9/2} := \sup_{|w - s_{\mathrm{ref}}| = 9/2} |\tilde{\xi}_{\mathrm{DH}}(w)| \le 1.123 \times 10^8$ (Appendix B). Along the trajectory from $h = 0$ to $h = \frac{311}{500}$, $|s(z_k) - s_{\mathrm{ref}}| < \frac{1}{4}$, so $\tilde{\xi}_{\mathrm{DH}}(s(z_k) + w)$ is analytic on $|w| \le R_k := \frac{9}{2} - |s(z_k) - s_{\mathrm{ref}}| > \frac{17}{4}$ with Cauchy coefficient bounds $|\tilde{\xi}_{\mathrm{DH}}^{(n)}(s(z_k))|/\Gamma(n+1) \le M_{9/2} R_k^{-n}$. Partitioning the entire compact interval $[0, \frac{311}{500}]$ into 400 overlapping interval balls $H_k = [h_k \pm \eta_k]$ (180 balls on $[0, \frac{285}{500}]$ with square-box radius $r_k = 6 \times 10^{-3}$, 120 balls on $[\frac{285}{500}, \frac{308}{500}]$ with $r_k = 2.5 \times 10^{-3}$, and 100 balls on $[\frac{308}{500}, \frac{311}{500}]$ with $r_k = 1.2 \times 10^{-3}$, with $0 \in H_1$ and $\frac{311}{500} \in H_{400}$), our 320-bit verification script encloses $g_{\mathrm{sinc}}(z_k, h_k)$, $\partial_h g_{\mathrm{sinc}}(z_k, H_k)$, and $\partial_z g_{\mathrm{sinc}}(B_k, H_k)$ (via third-order Taylor expansion in $z - z_k$ with $\sup_{z \in B_k} |z - z_k| = \sqrt{2}\, r_k$ and fourth-derivative Cauchy remainder on $R_k - \frac{\sqrt{2}}{2} r_k - \sup|H_k| > \frac{7}{2}$) and verifies that

$$N(B_k, H_k) := z_k - \frac{g_{\mathrm{sinc}}(z_k, h_k) + (H_k - h_k)\,\partial_h g_{\mathrm{sinc}}(z_k, H_k)}{\partial_z g_{\mathrm{sinc}}(B_k, H_k)} \subset \operatorname{int}(B_k),$$
$$\operatorname{Im} N(B_k, H_k) < 0$$

across all 400 slices, together with $N(B_{k-1}, h_{k-1,k}) \subset B_k$ at each overlap midpoint $h_{k-1,k} \in H_{k-1} \cap H_k$ ($2 \le k \le 400$) so uniqueness in B_k glues the slices into the single continuous zero branch originating from ρ_{2442} at $h = 0$. At $h = \frac{311}{500}$, evaluating the 64-term Taylor series at z_{sinc} (remainder $\le M_{9/2}(h/R_{\mathrm{pt}})^{64}/(1 - h/R_{\mathrm{pt}}) < 10^{-46}$) and the difference quotient $\partial_z g_{\mathrm{sinc}}(B_{\mathrm{sinc}}, \frac{311}{500}) = \frac{\mathrm{i}}{4h}[\tilde{\xi}_{\mathrm{DH}}(s(B_{\mathrm{sinc}}) + h) - \tilde{\xi}_{\mathrm{DH}}(s(B_{\mathrm{sinc}}) - h)]$ certifies $B_{\mathrm{sinc}} \subset B_{400}$, $N(B_{\mathrm{sinc}}) \subset \operatorname{int}(B_{\mathrm{sinc}})$, and $\operatorname{Im} N(B_{\mathrm{sinc}}) < 0$, proving (3.4) and $h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) > \frac{311}{500}$ (Figure 1).

(ii) Fix $h \ge 31$. By evenness, conjugation symmetry, and Lemma 2.1(i), every zero of $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta)$ satisfies $|\delta| \le \Delta := \sigma^* - \frac{1}{2} < \frac{5}{8}$, and it suffices to prove $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) \ne 0$ for all $\gamma \ge 0$ and all $\delta \in (0, \Delta]$. Substituting $v = \delta + u$ in $\int_{-h}^{h} \xi_{\mathrm{DH}}(\frac{1}{2} + \delta + u + \mathrm{i}\gamma)\,\mathrm{d}u$ and splitting at $v = 0$ via $\xi_{\mathrm{DH}}(\frac{1}{2} - v + \mathrm{i}\gamma) = \overline{\xi_{\mathrm{DH}}(\frac{1}{2} + v + \mathrm{i}\gamma)}$ gives

$$\begin{aligned} 2h\, G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) &= J(\gamma, h + \delta) + \overline{J(\gamma, h - \delta)}, \\ J(\gamma, Y) &:= \int_0^Y \xi_{\mathrm{DH}}\Big(\tfrac{1}{2} + v + \mathrm{i}\gamma\Big)\,\mathrm{d}v. \end{aligned} \tag{3.6}$$

We show that $\operatorname{Re}(J(\gamma,Y)/\partial_Y J(\gamma,Y)) > 0$ for all $Y \ge 30$ and all $\gamma \ge 0$. Set $s_+ := \frac{1}{2} + Y + \mathrm{i}\gamma$ ($\partial_Y J(\gamma,Y) = \xi_{\mathrm{DH}}(s_+) \ne 0$) and $z_+ := \frac{s_+ + 1}{2} = \frac{3}{4} + \frac{Y}{2} + \frac{\mathrm{i}\gamma}{2}$, so $\operatorname{Re} z_+ \ge \frac{63}{4} = 15.75$, $|z_+| \ge 15.75$, and $\alpha := \frac{1}{2}\log\frac{5 z_+}{\pi} = \alpha_R + \mathrm{i}\alpha_I$ satisfies $\alpha_R \ge \frac{1}{2}\log\frac{5 \times 15.75}{\pi} > 1.61$ and $\alpha_I \in [0, \frac{\pi}{4})$. Substitute $v = Y - w$ ($w \in [0, Y]$) and split $[0, Y] = [0, 6] \cup [6, Y - 4] \cup [Y - 4, Y]$:

(a) On $w \in [0, 6]$, $\operatorname{Re}(s_+ - w) \ge 24.5$ gives $|\mathcal{D}_{\mathrm{DH}}(s_+ - w)/\mathcal{D}_{\mathrm{DH}}(s_+) - 1| < 10^{-6}$, while Binet's first integral $|\psi(z) - \log z + \frac{1}{2z}| \le \frac{1}{12(\operatorname{Re} z)^2}$ on $\operatorname{Re}(z_+ - t) \ge 12.75$ ($t \in [0, 3]$) gives

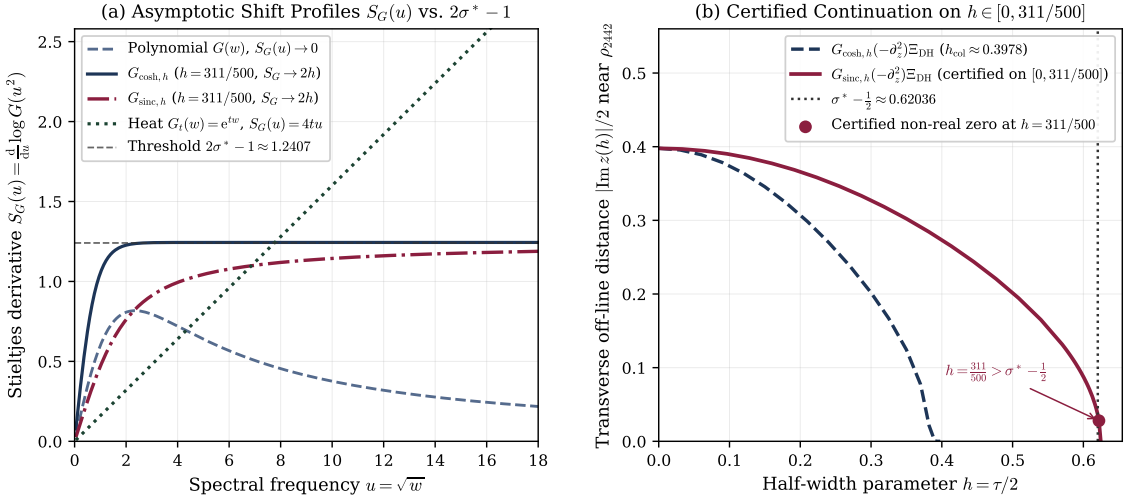


FIGURE 1. (a) Stieltjes logarithmic-derivative profiles $S_G(u)$ relative to the threshold $2\sigma^*-1$. (b) Transverse off-line ordinate $|\mathrm{Im}\, z(h)|/2$ near ρ_{2442} under $G_{\cosh,h}$ versus $G_{\mathrm{sinc},h}$, showing the 400-ball singularity-free continuation across $h\in[0,311/500]$ past $\sigma^*-\frac{1}{2}$.

$|\log\frac{\Gamma_{\mathrm{DH}}(s_+-w)}{\Gamma_{\mathrm{DH}}(s_+)}+\alpha w|\le\frac{0.20(w^2+2.1w)}{|z_+|}<0.49$, so $|\frac{\Gamma_{\mathrm{DH}}(s_+-w)}{\Gamma_{\mathrm{DH}}(s_+)}-e^{-\alpha w}|\le\frac{0.33(w^2+2.1w)}{|z_+|}e^{-\alpha_R w}$. Since $\mathrm{Re}(\frac{1}{\alpha})\ge\frac{1}{\alpha_R(1+\frac{\pi^2}{16\alpha_R^2})}>\frac{0.807}{\alpha_R}$ and $e^{-6\alpha_R}<10^{-4}$, the $[0,6]$ integral has real part $>\frac{0.76}{\alpha_R}$.

(b) For all z with $\mathrm{Re}\, z>0$, Binet's first integral bounds $|\log\Gamma(z)-(z-\frac{1}{2})\log z+z-\frac{1}{2}\log(2\pi)|\le\frac{1}{12\,\mathrm{Re}\, z}$. Setting $g(x):=\frac{x}{2}\log\frac{\pi}{5}+\mathrm{Re}[(z_+-\frac{x}{2}-\frac{1}{2})\log(z_+-\frac{x}{2})-(z_+-\frac{x}{2})]$, we have $g'(x)=-\frac{1}{2}\log(\frac{5}{\pi}|z_+-\frac{x}{2}|)+\frac{\mathrm{Re}\, z_+-x/2}{4|z_+-x/2|^2}$, which satisfies $g(6)-g(0)\le-3\log(\frac{5|z_+|}{\pi})+0.78$, $g'(x)<-0.64$ on $x\in[6,Y-4]$ ($\mathrm{Re}\, z_+-\frac{x}{2}\ge 2.75$), and $g'(x)<+0.25$ on $x\in[Y-4,Y]$ ($\mathrm{Re}\, z_+-\frac{x}{2}\ge\frac{3}{4}$). Including the Binet remainder <0.13 yields $|\frac{\Gamma_{\mathrm{DH}}(s_+-w)}{\Gamma_{\mathrm{DH}}(s_+)}|<2.5(\frac{5|z_+|}{\pi})^{-3}e^{-0.64(w-6)}$ on $[6,Y-4]$ (where $|\mathcal{D}_{\mathrm{DH}}(s_+-w)|\le\zeta(4.5)<1.06$, contributing $<\frac{0.001}{\alpha_R}$) and $<2.5e^{-11.8}(\frac{5|z_+|}{\pi})^{-3}$ on $[Y-4,Y]$ (where Lemma 4.2 bounds $|\mathcal{D}_{\mathrm{DH}}(s_+-w)|\le\max(38,4\gamma^{1/2})\le 38|z_+|^{1/2}$, contributing $<\frac{10^{-5}}{\alpha_R}$).

Combining (a) and (b) gives $\mathrm{Re}(J(\gamma,Y)/\partial_Y J(\gamma,Y))>\frac{0.70}{\alpha_R}>0$, whence $J(\gamma,Y)\neq 0$ and $\frac{\partial}{\partial Y}\log|J(\gamma,Y)|=\mathrm{Re}(\partial_Y J/J)>0$ for all $Y\ge 30$. Since $h\ge 31$ and $\delta\in(0,\Delta]$ imply $h+\delta>h-\delta\ge 31-\frac{5}{8}>30$, strict monotonicity gives $|J(\gamma,h+\delta)|>|J(\gamma,h-\delta)|=|\overline{J(\gamma,h-\delta)}|$ in (3.6), so $G_{\mathrm{sinc},h}(-\partial_z^2)\Xi_{\mathrm{DH}}(2\gamma-2i\delta)\neq 0$. □

4. Gauss–Weierstrass Asymptotics and the Ki–Kim–Lee Upper Bound

In this section we adapt the method of Ki, Kim, and Lee [13] (who proved $\Lambda_\zeta<1/2$ by combining large-height zero simplicity of H_t with de Bruijn's hyperbola inequality [8]) to $\Xi_{\mathrm{DH}}(z)$. Writing the Gaussian convolution $H_t^{\mathrm{DH}}(z)=(4\pi t)^{-1/2}\int_{-\infty}^{\infty}e^{-w^2/(4t)}\Xi_{\mathrm{DH}}(z-iw)\,dw$ ($t>0$) at $z=2\gamma-2i\delta$ via $v=\delta+w/2$, splitting $\mathbb{R}=[0,\infty)\cup(-\infty,0)$, and applying

$\xi_{\mathrm{DH}}(\frac12 - v + \mathrm{i}\gamma) = \overline{\xi_{\mathrm{DH}}(\frac12 + v + \mathrm{i}\gamma)}$ for real γ, v gives

$$\begin{aligned} H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) &= A_t(\gamma,\delta) + \overline{A_t(\gamma,-\delta)}, \\ A_t(\gamma,\delta) &:= \frac{1}{\sqrt{\pi t}} \int_0^\infty \mathrm{e}^{-(v-\delta)^2/t}\, \xi_{\mathrm{DH}}\Big(\tfrac12 + v + \mathrm{i}\gamma\Big)\, \mathrm{d}v, \\ \partial_\gamma A_t(\gamma,\delta) &= \mathrm{i}\, \partial_\delta A_t(\gamma,\delta) - \frac{\mathrm{i}}{\sqrt{\pi t}}\, \mathrm{e}^{-\delta^2/t}\, \xi_{\mathrm{DH}}\Big(\tfrac12 + \mathrm{i}\gamma\Big), \end{aligned} \tag{4.1}$$

where the third relation follows on integrating $\partial_\delta(\mathrm{e}^{-(v-\delta)^2/t}) = -\partial_v(\mathrm{e}^{-(v-\delta)^2/t})$ by parts over $[0,\infty)$, the boundary contribution at $+\infty$ vanishing because $|\xi_{\mathrm{DH}}(\frac12 + v + \mathrm{i}\gamma)| \ll_\gamma (1+v)^C \mathrm{e}^{\frac{v}{2}\log(v+2)}$ is dominated by $\mathrm{e}^{-(v-\delta)^2/t}$.

Lemma 4.1 (de Bruijn [8, Thms. 12 and 13])**.** *Let $F(z)$ be a real-even entire function of genus 0 or 1 whose zeros all satisfy $|\operatorname{Im} z| \le \Delta_0$.*

(i) *For every $t > 0$, all zeros of $\mathrm{e}^{-t\partial_z^2}F(z)$ satisfy $|\operatorname{Im} z| \le \sqrt{\max(0, \Delta_0^2 - 2t)}$; in particular, $\Lambda_{\mathrm{DH}} \le 2(\sigma^* - \frac12)^2$.*

(ii) *If $x_* \in \mathbb{R}$ and $\widetilde{\Delta} \ge 0$ are such that every zero z_ρ of F satisfies $(\operatorname{Im} z_\rho)^2 - (\operatorname{Re} z_\rho - x_*)^2 \le \widetilde{\Delta}^2$, then every zero z of $\mathrm{e}^{-t\partial_z^2}F(z)$ on the vertical line $\operatorname{Re} z = x_*$ satisfies $(\operatorname{Im} z)^2 \le \max(0, \widetilde{\Delta}^2 - 2t)$.*

We next state the explicit Dirichlet bounds proved in Appendix A.

Lemma 4.2 (Explicit Abel and Dirichlet-tail bounds)**.** *For all $v \ge 0$, $|\mathcal{D}_{\mathrm{DH}}(\frac12 + v + \mathrm{i}\tau)| \le 4|\tau|^{1/2}$ when $|\tau| \ge 100$, and $|\mathcal{D}_{\mathrm{DH}}(\frac12 + v + \mathrm{i}\tau)| \le 38$ when $|\tau| \le 100$. Moreover, for all $\tau \in \mathbb{R}$ and $u \ge \frac54$, $\sum_{n=2}^\infty |a_{\mathrm{DH}}(n)| n^{-1/2-u} < \frac12$, so $\operatorname{Re}\mathcal{D}_{\mathrm{DH}}(\frac12 + u + \mathrm{i}\tau) > \frac12$ and $|\mathcal{D}_{\mathrm{DH}}(\frac12 + u + \mathrm{i}\tau)| < \frac32$.*

Proposition 4.3 (Explicit height cutoff for H_t^{DH}; cf. Ki–Kim–Lee [13])**.** *Fix $t > 0$ and set $\Delta := \sigma^* - \frac12 < \frac58$. Define the explicit cutoff height*

$$\gamma_0(t) := \max\left(400 + 64t^3,\ \frac{2\pi}{5} \exp\left(\frac{4}{t}\Big(\Delta + 5 + 6\sqrt{t} + 2t\Big)\right)\right). \tag{4.2}$$

Then for all $|\gamma| \ge \gamma_0(t)$, $H_t^{\mathrm{DH}}(2\gamma - 2\mathrm{i}\delta) \ne 0$ for every $\delta \ne 0$, and every real zero of $H_t^{\mathrm{DH}}(x)$ in $|x| \ge 2\gamma_0(t)$ is simple.

Proof. By evenness and Lemma 4.1(i), it suffices to consider $\gamma \ge \gamma_0(t)$ and $|\delta| \le \Delta$. In (4.1), fix any $\eta \in [-\Delta, \Delta]$ and deform $v \in [0,\infty)$ to $\mathcal{V}_\theta \cup \mathcal{C}_\theta$, where $\mathcal{V}_\theta := \{\mathrm{i}y : 0 \le y \le \theta\}$, $\mathcal{C}_\theta := \{u + \mathrm{i}\theta : u \ge 0\}$, and $\theta := \frac{\pi t}{8}$; Cauchy's theorem applies because ξ_{DH} is entire and $|\mathrm{e}^{-(u+\mathrm{i}y-\eta)^2/t}\xi_{\mathrm{DH}}(\frac12 + u + \mathrm{i}(\gamma + y))| \ll_{\gamma,\theta} \mathrm{e}^{-u^2/(2t)} \to 0$ as $u \to +\infty$ uniformly for $y \in [0,\theta]$. Set $\tilde\gamma := \gamma + \theta > \gamma$ and $K_0(\gamma,\eta) := \mathrm{e}^{\frac{\pi^2 t}{64} + \mathrm{i}\frac{\pi}{4}\eta}\Gamma_{\mathrm{DH}}(\frac12 + \mathrm{i}\tilde\gamma) \ne 0$. Along $v = u + \mathrm{i}\theta$, expanding $-(u + \mathrm{i}\theta - \eta)^2/t = -\frac{(u-\eta)^2}{t} + \frac{\pi^2 t}{64} - \mathrm{i}\frac{\pi}{4}(u - \eta)$ cancels the $+\mathrm{i}\frac{\pi}{4}u$ Stirling phase of $\Gamma_{\mathrm{DH}}(\frac12 + u + \mathrm{i}\tilde\gamma)/\Gamma_{\mathrm{DH}}(\frac12 + \mathrm{i}\tilde\gamma) = (\frac{5\tilde\gamma}{2\pi})^{u/2}\mathrm{e}^{\mathrm{i}\frac{\pi}{4}u + E_{\mathrm{St}}(u,\tilde\gamma)}$, giving

$$\frac{\sqrt{\pi t}\, \mathrm{e}^{-u_0^2/t + \eta^2/t}}{K_0(\gamma,\eta)}\, A_t(\gamma,\eta) = \int_0^\infty \mathrm{e}^{-(u-u_0)^2/t + E_{\mathrm{St}}(u,\tilde\gamma)}\, \mathcal{D}_{\mathrm{DH}}\Big(\tfrac12 + u + \mathrm{i}\tilde\gamma\Big)\, \mathrm{d}u + \mathcal{E}_{\mathcal{V}}(\gamma,\eta), \tag{4.3}$$

where $u_0 = u_0(\gamma,\eta) := \eta + \frac{t}{4}\log\frac{5\tilde\gamma}{2\pi} > 5 + 6\sqrt{t} + 2t$. As shown in Appendix A, $|E_{\mathrm{St}}(u,\tilde\gamma)| \le \frac{u^2+6u}{4\tilde\gamma}$ for all $u \ge 0$, so $|E_{\mathrm{St}}(u,\tilde\gamma)| < \frac{1}{10}$ on $|u - u_0| \le 3\sqrt{t}$ (where $u > 5 > \frac54$ and $\operatorname{Re}(\mathrm{e}^{E_{\mathrm{St}}}\mathcal{D}_{\mathrm{DH}}) \ge \mathrm{e}^{-1/10}(\frac12\cos\frac{1}{10} - \frac32\sin\frac{1}{10}) > 0.31$), while $[0,\frac54] \cup [\frac54, u_0 - 3\sqrt{t}]$, $\mathcal{V}_\theta$, and $u \ge u_0 + 3\sqrt{t}$ contribute less than $0.015\sqrt{\pi t}$ in modulus, leaving real part $> 0.29\sqrt{\pi t}$. Writing $\frac{2}{t}(v - \eta) = (\frac12\log\frac{5\tilde\gamma}{2\pi} + \mathrm{i}\frac{\pi}{4}) + \frac{2}{t}(u - u_0)$ in $\partial_\eta A_t(\gamma,\eta)$ bounds the centered-moment remainder by $\frac{6.7}{\sqrt{t}} < \frac14\log\frac{5\tilde\gamma}{2\pi}$ (Appendix A), whence $A_t(\gamma,\eta) \ne 0$ and

$$\operatorname{Re}\left(\frac{\partial_\eta A_t(\gamma,\eta)}{A_t(\gamma,\eta)}\right) \ge \frac14\log\frac{5\tilde\gamma}{2\pi} > 1 \qquad \forall\, \eta \in [-\Delta,\Delta]. \tag{4.4}$$

Integrating (4.4) over $[-\delta,\delta]$ for $\delta\in(0,\Delta]$ gives $|A_t(\gamma,\delta)|>|A_t(\gamma,-\delta)|$ and hence $H_t^{\mathrm{DH}}(2\gamma-2\mathrm{i}\delta)\neq 0$. At $\delta=0$, $H_t^{\mathrm{DH}}(2\gamma)=2|A_t(\gamma,0)|\cos\Theta_t(\gamma)$ with $\Theta_t(\gamma):=\arg A_t(\gamma,0)$, and the third identity of (4.1) together with the boundary ratio bound $<10^{-2}$ from Appendix A gives

$$\Theta_t'(\gamma)=\mathrm{Re}\biggl(\frac{\partial_\delta A_t(\gamma,0)}{A_t(\gamma,0)}\biggr)-\mathrm{Re}\biggl(\frac{\xi_{\mathrm{DH}}(\frac12+\mathrm{i}\gamma)}{\sqrt{\pi t}\,A_t(\gamma,0)}\biggr)\geq\frac14\log\frac{5\gamma}{2\pi}-10^{-2}>1>0, \tag{4.5}$$

so $\frac{\mathrm{d}}{\mathrm{d}\gamma}H_t^{\mathrm{DH}}(2\gamma)=\mp 2|A_t(\gamma,0)|\Theta_t'(\gamma)\neq 0$ at every real zero $\cos\Theta_t(\gamma)=0$. □

Applying the finite-box hyperbola argument of Ki, Kim, and Lee [13, §3] and de Bruijn [8, Thm. 12] immediately sharpens Lemma 4.1(i) to a strict inequality.

Corollary 4.4 (Strict upper bound for Λ_{DH}; after Ki–Kim–Lee [13])**.** *The de Bruijn–Newman constant of $\Xi_{\mathrm{DH}}(z)$ satisfies*

$$\Lambda_{\mathrm{DH}}<2\Bigl(\sigma^*-\tfrac12\Bigr)^2=0.76969925830451198120\ldots \tag{4.6}$$

Proof. Set $\Delta_0:=2(\sigma^*-\frac12)$ and $t_1:=\frac12<\frac12\Delta_0^2=2(\sigma^*-\frac12)^2$. By Proposition 4.3, $H_{t_1}^{\mathrm{DH}}(z)$ has only real zeros in $|\mathrm{Re}\,z|\geq 2\gamma_0(t_1)$ and hence at most finitely many non-real zeros overall. If $H_{t_1}^{\mathrm{DH}}$ has no non-real zeros, then $\Lambda_{\mathrm{DH}}\leq t_1<2(\sigma^*-\frac12)^2$. Otherwise, $Y_1:=\max\{|\mathrm{Im}\,w|:H_{t_1}^{\mathrm{DH}}(w)=0\}>0$ is attained at some $w_*=x_*+\mathrm{i}y_*$ with $|x_*|<2\gamma_0(t_1)$. By Lemma 3.1, every zero z_ρ of $\Xi_{\mathrm{DH}}(z)$ satisfies $|\mathrm{Im}\,z_\rho|<\Delta_0$ strictly. Setting $R:=2\gamma_0(t_1)+\Delta_0+1$, the finitely many zeros of Ξ_{DH} in $|\mathrm{Re}\,z_\rho|\leq R$ satisfy $\Delta_R:=\max_{|\mathrm{Re}\,z_\rho|\leq R}|\mathrm{Im}\,z_\rho|<\Delta_0$, while zeros with $|\mathrm{Re}\,z_\rho|>R$ satisfy $|\mathrm{Re}\,z_\rho-x_*|>\Delta_0+1>|\mathrm{Im}\,z_\rho|$, so $(\mathrm{Im}\,z_\rho)^2-(\mathrm{Re}\,z_\rho-x_*)^2\leq\Delta_R^2<\Delta_0^2$ globally. Lemma 4.1(ii) at $\mathrm{Re}\,z=x_*$ yields $Y_1^2\leq\max(0,\Delta_R^2-2t_1)$, and Lemma 4.1(i) applied to $H_{t_1+Y_1^2/2}^{\mathrm{DH}}=\mathrm{e}^{-(Y_1^2/2)\partial_z^2}H_{t_1}^{\mathrm{DH}}$ gives $\Lambda_{\mathrm{DH}}\leq t_1+\frac12Y_1^2\leq\max(t_1,\frac12\Delta_R^2)<\frac12\Delta_0^2=2(\sigma^*-\frac12)^2$. □

5. Certified Lower Bound on Λ_{DH} via Ball Arithmetic

5.1. **Rigorous theta-kernel truncation and Interval–Newton certificate.** To certify an unconditional lower bound on Λ_{DH} free of quadrature aliasing or root-tracking heuristics, we enclose $H_t^{\mathrm{DH}}(z)=2\int_0^\infty\mathrm{e}^{tu^2}\Phi_{\mathrm{DH}}(u)\cos(zu)\,\mathrm{d}u$ and its first three z-derivatives in complex ball arithmetic [11] on $[0,U]$ with analytical remainder bounds.

Lemma 5.1 (Theta-kernel truncation and fourth-derivative enclosure)**.** *Let $0\leq t\leq\frac14$, $2\leq U\leq\frac52$, $N\geq 25$, and $|y|\leq 1$, where $y=\mathrm{Im}\,z$. Approximating $\Phi_{\mathrm{DH}}(u)$ on $[0,U]$ by $\Phi_{\mathrm{DH}}^{(<N)}(u):=4\mathrm{e}^{3u}\sum_{n=1}^{N-1}n\,a_{\mathrm{DH}}(n)\,\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}/5}$ incurs a truncation error bounded for $k\in\{0,1,2,3,4\}$ by*

$$\biggl|\partial_z^kH_t^{\mathrm{DH}}(z)-2\int_0^U\mathrm{e}^{tu^2}\Phi_{\mathrm{DH}}^{(<N)}(u)\,\partial_z^k\cos(zu)\,\mathrm{d}u\biggr|\leq E_1(U,N,k,|y|)+E_2(U), \tag{5.1}$$

where $r_N:=\mathrm{e}^{-\pi N/5}$, $E_1(U,N,k,|y|):=8U^{k+1}\mathrm{e}^{tU^2+(3+|y|)U}\frac{r_N^N(N-(N-1)r_N)}{(1-r_N)^2}$, and $E_2(U):=\frac{40}{\pi}\mathrm{e}^{-4U-\frac{\pi}{10}\mathrm{e}^{4U}}$. Moreover, on any closed box B with $\max_{z\in B}|\mathrm{Im}\,z|\leq y_{\max}\leq 1$, one has

$$\sup_{z\in B}\bigl|\partial_z^4H_t^{\mathrm{DH}}(z)\bigr|\leq 4N(N-1)U^5\mathrm{e}^{tU^2+3U}\cosh(Uy_{\max})+E_1(U,N,4,y_{\max})+E_2(U)=:M_4(B). \tag{5.2}$$

The proof of Lemma 5.1 is given in Appendix B. At $z_0\approx 480.69-0.00384\,\mathrm{i}$ ($\gamma\approx 240.35$), $|\partial_zH_t^{\mathrm{DH}}(z_0)|\approx 3\times10^{-84}$ due to $\mathrm{e}^{-\pi\gamma/4}\approx 10^{-82}$ oscillatory decay; expanding $\partial_zH_t^{\mathrm{DH}}(z)$ on $B(z_0,r):=\{z\in\mathbb{C}:|\mathrm{Re}(z-z_0)|\leq r,\ |\mathrm{Im}(z-z_0)|\leq r\}$ to third order via $\partial_zH_t^{\mathrm{DH}}(z_0)+(z-z_0)\partial_z^2H_t^{\mathrm{DH}}(z_0)+\frac12(z-z_0)^2\partial_z^3H_t^{\mathrm{DH}}(z_0)$ with Taylor remainder bounded by $\frac16|z-z_0|^3M_4(B)\leq\frac{\sqrt2}{3}r^3M_4(B)$ allows rigorous certification at sober 448-bit working precision with box radius $r=10^{-32}$.

TABLE 1. Summary of `FLINT/Arb` ball-arithmetic enclosures verified by `verify_paper15_debruijn_newman_dh.py`.

Target / Witness	Param.	Certified Midpoint / Cover (16 digits shown)	Radius r
Supremum σ^* (320-bit)	IVT	1.1203624981833251	10^{-30}
G_{sinc} cover (320-bit)	$[0, \frac{311}{500}]$	400 overlapping balls H_k, $\operatorname{Im} N(B_k, H_k) < 0$	$\le 6 \times 10^{-3}$
G_{sinc} (ρ_{2442}, 320-bit)	$h = \frac{311}{500}$	$4884.886654191757 - 0.0560046288608581\,\mathrm{i}$	10^{-30}
H_t^{DH} (ρ_{240}, 448-bit)	$t = \frac{1153}{5000}$	$480.6902350136200 - 0.0038371695655869\,\mathrm{i}$	10^{-32}

Theorem 5.2 (Interval-certified lower bound on Λ_{DH})**.** *At exact rational* $t = \frac{1153}{5000} = 0.2306$*, the entire function* $H^{\mathrm{DH}}_{1153/5000}(z)$ *possesses a certified simple non-real zero inside* $B(z_{240}, 10^{-32})$ *centered at*

$$\begin{aligned} z_{240} = {} & 480.690235013620019064551671086167379 \\ & - 0.003837169565586910716957555909571243\,\mathrm{i}, \end{aligned} \tag{5.3}$$

originating from the off-line zero $\rho_{240} = 0.8695305796\ldots + 240.4046723514\ldots\,\mathrm{i}$ *of* $\mathcal{D}_{\mathrm{DH}}(s)$*. Consequently,* $\Lambda_{\mathrm{DH}} > \frac{1153}{5000} = 0.2306$*.*

Proof. Evaluating $\partial_z^k H^{\mathrm{DH}}_{1153/5000}(z_{240})$ ($k \in \{0,1,2,3\}$) with $U = \frac{57}{25} = 2.28$ and $N = 32$ via 448-bit `FLINT/Arb` Gauss–Legendre ball quadrature (`acb.integral` [11]) plus the analytical remainder balls of Lemma 5.1, and enclosing $\partial_z H^{\mathrm{DH}}_{1153/5000}(B)$ on $B := B(z_{240}, 10^{-32})$ via (5.2), our verification script confirms that $0 \notin \partial_z H^{\mathrm{DH}}_{1153/5000}(B)$, $N(B) := z_{240} - H^{\mathrm{DH}}_{1153/5000}(z_{240})/\partial_z H^{\mathrm{DH}}_{1153/5000}(B) \subset \operatorname{int}(B)$, and $\sup_{w \in N(B)} \operatorname{Im} w < -3.83 \times 10^{-3} < 0$ (Table 1). By Brouwer's fixed-point theorem and the Argument Principle, $H^{\mathrm{DH}}_{1153/5000}$ has a unique simple zero in B, which is strictly non-real. If $H^{\mathrm{DH}}_{t_0}$ had only real zeros for some $t_0 \le \frac{1153}{5000}$, Lemma 4.1(i) applied to $H^{\mathrm{DH}}_{1153/5000} = \mathrm{e}^{-(1153/5000 - t_0)\partial_z^2} H^{\mathrm{DH}}_{t_0}$ would force all zeros at $t = \frac{1153}{5000}$ to be real, a contradiction; openness of simple non-real zeros in t yields $\Lambda_{\mathrm{DH}} > \frac{1153}{5000}$. □

Remark 5.3. Combining Theorem 5.2 with Platt and Trudgian's unconditional upper bound $\Lambda_\zeta \le \frac{1}{5} = 0.2$ [18] separates the two constants strictly under the common normalization of Section 1.2: $\Lambda_\zeta \le \frac{1}{5} < \frac{1153}{5000} < \Lambda_{\mathrm{DH}}$.

5.2. **Numerical collision survey versus certified theorems.** We strictly distinguish the ball-certified theorems in Table 1 from exploratory numerical root tracking:

(1) *Oscillation aliasing in non-interval quadrature:* Near the first off-line zero $\rho_1 \approx 0.808517 + 85.699348\,\mathrm{i}$ ($z \approx 171.40 - 0.6170\,\mathrm{i}$), $\cos(zu)$ undergoes roughly 60 oscillations on $[0, 2.25]$; uncertified adaptive quadrature (`mpmath.quad` at default subdivision) aliases across sign changes and reports a spurious coalescence near $t \approx 0.1419$, whereas high-precision scalar Newton refinement places coalescence at $t_{\mathrm{col}}(\rho_1) \approx 0.17650538$.

(2) *High-height collision survey and leading-order heuristic:* High-precision scalar Newton refinement on $(H_t^{\mathrm{DH}}, \partial_z H_t^{\mathrm{DH}}) = (0,0)$ locates the coalescence of ρ_{240} at $t_{\mathrm{col}}(\rho_{240}) \approx 0.2306073618964546$ ($x_{\mathrm{col}} \approx 480.6902310425$) and that of ρ_{531} at $t_{\mathrm{col}}(\rho_{531}) \approx 0.20127583$. An exploratory scan of off-line zeros of $\mathcal{D}_{\mathrm{DH}}(s)$ up to $\gamma = 72{,}500$ (Figure 2(b)) detects no collision exceeding $t_{\mathrm{col}}(\rho_{240})$: although deeper real parts occur at higher ordinates (such as ρ_{2442} with $\operatorname{Re}\rho \approx 0.89775$ and $t_{\mathrm{col}} \approx 0.19973$, or $\gamma \approx 49{,}688.34$ with $\operatorname{Re}\rho \approx 0.93187$ and $t_{\mathrm{col}} \approx 0.1609$), matching the Gaussian saddle $u_0 \approx \frac{t}{4}\log\frac{5\gamma}{2\pi}$ of Section 4 to $\sigma^* - \frac{1}{2}$ suggests the unproved leading-order heuristic $t_{\mathrm{col}}(\rho) \lesssim 4(\sigma^* - \frac{1}{2})/\log(\frac{5\gamma}{2\pi})$ (note that Proposition 4.3 itself yields only the weaker rigorous ceiling $4(\sigma^* - \frac{1}{2} + 5 + 6\sqrt{t} + 2t)/\log(\frac{5\gamma}{2\pi})$). Whether Λ_{DH} coincides with $t_{\mathrm{col}}(\rho_{240})$ remains an open finite-box verification problem.

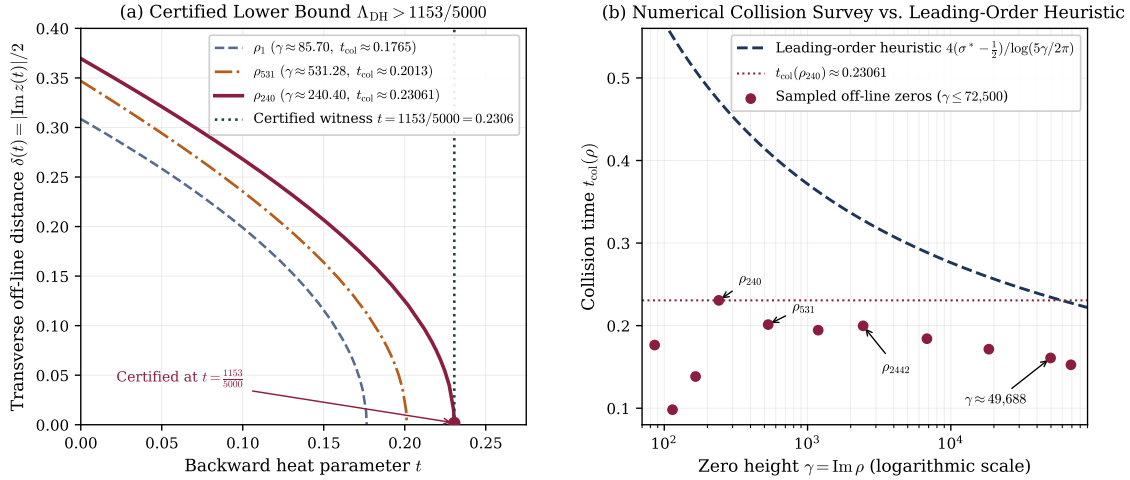


FIGURE 2. (a) Transverse off-line distance $\delta(t) = |\operatorname{Im} z(t)|/2$ under backward heat flow for ρ_1, ρ_{531}, and ρ_{240}, with the certified Interval–Newton witness at $t = 1153/5000 = 0.2306$. (b) Exploratory numerical collision survey up to $\gamma = 72{,}500$ compared with the leading-order heuristic $4(\sigma^* - \frac{1}{2})/\log(\frac{5\gamma}{2\pi})$.

6. Scope, Limitations, and Open Questions

Theorem 1.1 shows that for self-dual Dirichlet combinations without an Euler product, real entire Pólya–Schur multipliers exhibit both an infinite-height shift threshold at $S_G(u) \to 2\sigma^* - 1$ (sensitive to root-gap regularity even when $\rho(G) \in (1/2, 1)$) and a strict gap between infinite-height real-rootedness and global real-rootedness for strip-averaging kernels such as $G_{\mathrm{sinc},h}$. Alongside the three open Stieltjes boundary regimes recorded in Remark 2.8, narrowing the certified intervals $\frac{311}{500} < h^*_{\mathrm{global}}(G_{\mathrm{sinc}}; \Xi_{\mathrm{DH}}) \le 31$ and $\frac{1153}{5000} < \Lambda_{\mathrm{DH}} < 2(\sigma^* - \frac{1}{2})^2$ reduces to finite two-dimensional argument-principle box covers once the height cutoffs in Theorem 3.2(ii) and Proposition 4.3 are sharpened via stationary-phase Dirichlet-polynomial approximations.

Appendix A. Proofs of Explicit Analytic Constants and Gaussian Tail Bounds

Proof of Lemma 4.2. Set $S(N) := \sum_{n=1}^{N} a_{\mathrm{DH}}(n)$. Because $a_{\mathrm{DH}}(n+5) = a_{\mathrm{DH}}(n)$ and $\sum_{r=1}^{5} a_{\mathrm{DH}}(r) = 1+\kappa-\kappa-1+0 = 0$, checking $N \equiv 1, 2, 3, 4, 0 \pmod 5$ gives $\max_{N \ge 1} |S(N)| = 1 + \kappa < 1.2841$. By conjugation symmetry $\mathcal{D}_{\mathrm{DH}}(\sigma - \mathrm{i}\tau) = \overline{\mathcal{D}_{\mathrm{DH}}(\sigma + \mathrm{i}\tau)}$, assume $\tau \ge 0$, set $\sigma := \frac{1}{2} + v \ge \frac{1}{2}$, and first consider $\tau \ge 100$ with $M := \lfloor \tau \rfloor \ge 100$. Splitting $\mathcal{D}_{\mathrm{DH}}(\sigma + \mathrm{i}\tau) = \sum_{n=1}^{M} a_{\mathrm{DH}}(n) n^{-\sigma - \mathrm{i}\tau} + \sum_{n=M+1}^{\infty} a_{\mathrm{DH}}(n) n^{-\sigma - \mathrm{i}\tau}$, the prefix satisfies

$$\begin{aligned} \left| \sum_{n=1}^{M} a_{\mathrm{DH}}(n) n^{-\sigma - \mathrm{i}\tau} \right| &\le 2(1+\kappa) \sum_{m=0}^{\lfloor (M-1)/5 \rfloor} (5m+1)^{-1/2} \\ &\le 2(1+\kappa)\Big(1 + \frac{2}{5}(\sqrt{\tau} - 1)\Big) \le 2(1+\kappa)(0.46\tau^{1/2}) < 1.19\, \tau^{1/2}. \end{aligned} \tag{A.1}$$

Summing $n \ge M+1$ by parts yields $-S(M)(M+1)^{-\sigma - \mathrm{i}\tau} + (\sigma + \mathrm{i}\tau) \int_{M+1}^{\infty} S(\lfloor x \rfloor) x^{-\sigma - 1 - \mathrm{i}\tau}\, \mathrm{d}x$. For $\sigma \le 19\tau^{1/2}/20$, $|\sigma + \mathrm{i}\tau| \le 1.005\tau$ and $\int_{M+1}^{\infty} x^{-3/2}\, \mathrm{d}x < 2\tau^{-1/2}$, bounding the tail by $(1+\kappa)\tau^{-1/2} + 2.01(1+\kappa)\tau^{1/2} < 2.71\tau^{1/2}$ and $|\mathcal{D}_{\mathrm{DH}}| < 3.90\tau^{1/2} < 4\tau^{1/2}$; for $\sigma > 19\tau^{1/2}/20 \ge \frac{19}{2}$, $|\mathcal{D}_{\mathrm{DH}}| \le \zeta(19/2) < 1.002 < 4\tau^{1/2}$. When $0 \le \tau \le 100$, the same split at $M = 100$ gives $< 11.82 + 0.13 + 25.80 < 38$ for $\sigma \le \frac{19}{2}$, and $\zeta(19/2) < 38$ for $\sigma > \frac{19}{2}$. Finally, for $u \ge \frac{5}{4}$ ($\sigma \ge \frac{7}{4}$), Hurwitz evaluation at $\sigma = \frac{7}{4}$ gives $\sum_{n=2}^{\infty} |a_{\mathrm{DH}}(n)| n^{-7/4} = 0.41770137\ldots < \frac{1}{2}$, whence $\operatorname{Re} \mathcal{D}_{\mathrm{DH}} > \frac{1}{2}$ and $|\mathcal{D}_{\mathrm{DH}}| < \frac{3}{2}$. □

Completion of Binet and tail bounds in Proposition 4.3. Write $z_0 := \frac{3}{4} + \frac{\mathrm{i}\tilde{\gamma}}{2}$, so $|z_0| > \frac{\tilde{\gamma}}{2} \geq 200$. In Binet's second integral $\psi(z) - \log z + \frac{1}{2z} = -\int_0^\infty \frac{2s\,\mathrm{d}s}{(s^2+z^2)(\mathrm{e}^{2\pi s}-1)}$ at $z = z_0 + w$ ($w \geq 0$), splitting at $s = |z|/\sqrt{2}$ (where $|s^2+z^2| \geq \frac{1}{2}|z|^2$ on $[0, |z|/\sqrt{2}]$ and $|s^2+z^2| \geq 2(\operatorname{Re} z)(\operatorname{Im} z) \geq 300$ on $(|z|/\sqrt{2}, \infty)$) yields $|\psi(z_0+w) - \log(z_0+w) + \frac{1}{2(z_0+w)}| \leq \frac{1}{5|z_0|^2}$, and integrating over $w \in [0, u/2]$ together with $|\log(1+\frac{w}{z_0})| \leq \frac{w}{|z_0|}$ and $|\log(1-\frac{3\mathrm{i}}{2\tilde{\gamma}})| \leq \frac{3}{2\tilde{\gamma}}$ gives $|E_{\mathrm{St}}(u,\tilde{\gamma})| \leq \frac{u^2+6u}{4\tilde{\gamma}}$ for all $u \geq 0$. Set $X := \log\frac{5\tilde{\gamma}}{2\pi}$. By (4.2), $\frac{t}{4}X > \Delta + 5 + 6\sqrt{t} + 2t$ and $\tilde{\gamma} > \max(64t^3, \mathrm{e}^X) \geq 16t^2\mathrm{e}^{X/3}$. Since $u_0 + 3\sqrt{t} \in (5, \frac{t}{2}X)$, on $u \in [0, u_0 + 3\sqrt{t}]$ we have $|E_{\mathrm{St}}(u,\tilde{\gamma})| < \frac{11}{1280}X^2\mathrm{e}^{-X/3} < 0.042 < \frac{1}{10}$. On $u \geq u_0 + 3\sqrt{t}$, $\frac{u^2+6u}{4\tilde{\gamma}} < \frac{1}{10} + \frac{(u-u_0)^2}{2t}$, so Lemma 4.2 bounds the right tail by $\frac{3}{2}\mathrm{e}^{1/10}\int_{3\sqrt{t}}^\infty \mathrm{e}^{-y^2/(2t)}\,\mathrm{d}y < 0.01\sqrt{\pi t}$. On $[\frac{5}{4}, u_0 - 3\sqrt{t}]$, Lemma 4.2 bounds the integral by $\frac{3}{2}\mathrm{e}^{1/10}\int_{3\sqrt{t}}^\infty \mathrm{e}^{-y^2/t}\,\mathrm{d}y < 10^{-4}\sqrt{\pi t}$. On $[0, \frac{5}{4}]$ and $\mathcal{V}_\theta$, $\frac{(u_0-5/4)^2}{t} > \frac{17}{32}X + \frac{2.1}{t} + \frac{6.75}{\sqrt{t}} + 5.4 + 0.6t$ and $|\mathcal{D}_{\mathrm{DH}}| \leq 4\tilde{\gamma}^{1/2} < 4.52\mathrm{e}^{X/2}$ bound the $[0, \frac{5}{4}]$ piece by $< 10^{-3}\sqrt{\pi t}$, $|\mathcal{E}_{\mathcal{V}}| < 10^{-3}\sqrt{\pi t}$, and $|\xi_{\mathrm{DH}}(\frac{1}{2}+\mathrm{i}\gamma)/(\sqrt{\pi t}\,A_t(\gamma,0))| < 10^{-2}$. Finally, integrating $\frac{2|u-u_0|}{t}\mathrm{e}^{-(u-u_0)^2/t}$ over $[\frac{5}{4}, u_0+3\sqrt{t}]$ contributes $< \frac{1.88}{\sqrt{t}}\sqrt{\pi t}$ (plus $< \frac{0.06}{\sqrt{t}}\sqrt{\pi t}$ from the tails), and dividing by $0.29\sqrt{\pi t}$ gives $< \frac{6.7}{\sqrt{t}} < \frac{1}{4}X$, proving (4.4) and (4.5). □

Appendix B. Ball-Arithmetic Truncation Enclosures

Proof of Lemma 5.1 and Cauchy bound in Theorem 3.2. For $n \geq N \geq 25$ and $u \in [0, U]$, $\mathrm{e}^{4u} \geq 1$ and $n^2 \geq Nn$ give $\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}/5} \leq r_N^n$ with $r_N = \mathrm{e}^{-\pi N/5} < 10^{-6}$. Since $|a_{\mathrm{DH}}(n)| \leq 1$, summing $\sum_{n=N}^\infty n\,r_N^n = \frac{r_N^N(N-(N-1)r_N)}{(1-r_N)^2}$ and using $|\partial_z^k\cos(zu)| \leq U^k\mathrm{e}^{|y|U}$ over $[0,U]$ gives $E_1(U,N,k,|y|)$. For $u \geq U \geq 2$ and $k \in \{0,1,2,3,4\}$, $\sum_{n=1}^\infty n\,\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}/5} \leq 2\mathrm{e}^{-\pi\mathrm{e}^{4u}/5}$ and $tu^2 + (3+|y|)u + k\log u \leq \frac{u^2}{4} + 8u \leq \frac{\pi}{10}\mathrm{e}^{4u} - 4u$ ($u \geq 2$), so the tail on $[U,\infty)$ is bounded by $16\mathrm{e}^{-4u-\frac{\pi}{10}\mathrm{e}^{4u}} \leq \frac{40}{\pi}\mathrm{e}^{-4U}\cdot\frac{2\pi}{5}\mathrm{e}^{4u}\mathrm{e}^{-\frac{\pi}{10}\mathrm{e}^{4u}}$, integrating to $E_2(U)$. Bounding $\sum_{n=1}^{N-1} n|a_{\mathrm{DH}}(n)|\,\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}/5} \leq \frac{N(N-1)}{2}$ and $|u^4\cos(zu)| \leq U^4\cosh(Uy_{\max})$ on $[0,U]$ yields (5.2). Finally, in Theorem 3.2(i), $\xi_{\mathrm{DH}}(s_{\mathrm{ref}}+w)$ is entire in w; covering $|w| = \frac{9}{2}$ by 512 overlapping interval arcs $W_j := \frac{9}{2}\mathrm{e}^{\mathrm{i}\Theta_j}$ ($\Theta_j = \frac{2\pi}{512}[(j+\frac{1}{2}) \pm 0.5001]$) in 320-bit `FLINT/Arb` certifies $M_{9/2} := \max_{0\leq j<512}\sup_{w\in W_j}|\tilde{\xi}_{\mathrm{DH}}(s_{\mathrm{ref}}+w)| \leq 1.123\times10^8$, whence Cauchy's integral formula on $|w-(s(z_k)-s_{\mathrm{ref}})| = R_k := \frac{9}{2} - |s(z_k)-s_{\mathrm{ref}}|$ (and on $R_k - \frac{\sqrt{2}}{2}r_k - \sup|H_k| > \frac{7}{2}$ for $\partial_z^4 g_{\mathrm{sinc}}$ on $B_k\times H_k$) bounds all Taylor remainders in (3.5) and its derivatives. □

Use of AI and computational verification

Portions of the literature exploration, asymptotic stress-testing, manuscript drafting, and `FLINT/Arb` verification scripting were conducted with the assistance of an AI research assistant (**Gemini**). Every analytical inequality, contour deformation, Möbius inversion identity, and interval-arithmetic enclosure reported in this paper was independently checked against symbolic and arbitrary-precision ball-arithmetic computations; all proof certificates are deterministically verified by the open-source script `scripts/verify_paper15_debruijn_newman_dh.py` using `python-flint` [11] (with optional figure regeneration via `--regenerate-figures`).

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